From detector signals to Δ(x,Q²): the reconstruction chain of the inclusive and coherent cos 2φ measurements

2026-08-24 · revised 2026-08-26 · companion to the cos 2φ projection report (money plots 5–7) and to plans/07 WP3 · computed with `polligen.reco`, `polligen.hfs` and `polligen.recopseudo` (`evgen/scripts/`: `reco_chain_figures.py`, `hfs_resolution.py`, `money_cos2phi_reco.py`, `money_cos2phi_coherent_reco.py`) on the PYTHIA 8 hadronic final state of `tools/pythia8` · revision history in Appendix A

The projection report presents the double-helicity-flip structure function Δ(x,Q²) as φ′ pseudo-data and as extracted xΔ points. This note is the chain behind them: what the detector records event by event in the inclusive and in the coherent intact-^{6}Li measurement, how the azimuth carrying the observable is built from the reconstructed four-vectors and the known spin direction, which kinematic reconstruction assigns (x, Q²), which estimator turns spin-sorted φ′ histograms into an amplitude, how the amplitude becomes Δ, and how well the chain closes at the reconstructed level (§7).

> **Three findings that change the analysis.** (1) Three of the four inclusive sweet-spot bins of money plot 5 sit at y = 0.010–0.026, where the electron method gives δy/y = 46–118% even with a 1.2% calorimeter: x must come from the hadronic final state. On the PYTHIA 8 final state with 50 MeV of calorimeter noise the Σ method gives δy/y = 0.32 / 0.22 / 0.29 / 0.15 at the four spots; the reconstructed bins keep 43–69% purity (52–76% once the hadronic scale is calibrated) and their amplitude sits 5–21% below the true-bin value uncalibrated, a response worth fitting rather than assuming (§7). (2) The single-fill cos 2φ′ fit measures the detector's own cos 2φ′ acceptance harmonic divided by P_{zz}: a 3% harmonic aligned with the vertical spin axis — the symmetry axis of the crossing angle and of the Roman-Pot cutout alike — fakes an amplitude four times the signal. The spin-state-sorted ratio of m = ±1-rich and m = 0-rich fills cancels any common acceptance bin by bin, turns relative-luminosity errors into a constant, and is 1.5× *more* precise. (3) In the coherent channel the anchored deformation term modulates the *recoil* azimuth $\varphi_t - \varphi_S$, not the electron azimuth of the projection; and the Roman-Pot aperture, measured with an intact ^{6}Li through the ePIC geometry, opens *horizontally*, fakes ⟨cos 2β⟩ = −0.77 against a physics $a_2 \approx 0.036$, and leaves 1.4% of the coherent recoils at 40.8 GeV/u, 5×10⁻⁵ at 50 and no accepted recoil at all in the binned window at 50 or 137.5: a low-energy programme (§4, §7).

1 · What is measured

Both measurements share the trigger object — a scattered electron in the central detector — and the spin bookkeeping of the ion beam; the coherent one adds the intact ^{6}Li in the far-forward spectrometers and an exclusivity selection. The observable is a *Fourier coefficient of a ratio* of spin-sorted azimuthal distributions; Δ follows from it through kinematic factors and the unpolarized structure functions of the same data.

RECORDED PER EVENT	INCLUSIVE E ^{6}LI → E′ X	COHERENT E ^{6}LI → E′ X ^{6}LI(G.S.)
Beam and spin	Bunch-crossing ID → spin state (m = ±1-rich / m = 0-rich pattern); P_{zz} per fill from the polarimeter (scale ≈ 3%); alignment axis n̂, vertical at IP6, with its measured tilt; luminosity per spin state.	
Central detector	e′ candidate: track (p, θ, φ at the vertex) matched to an EMCal cluster, with E/p and shower shape for electron ID; the hadronic final state (all other tracks and clusters) for $\Sigma = \Sigma_h(E_h - p_{z,h})$; vertex.	The same, plus the diffractive system X (its mass M_X and the rapidity gap) — the only handle on x_P, since the recoil momentum is unresolved.
Far forward	Not required (breakup fragments of ^{6}Li are mostly beam-blind at IP6; plans/00).	Roman-Pot track at beam rigidity (|R − 1| < 0.05) with transverse displacement above the 10σ envelope (§4 step 1); vetoes in OMD (p, ^{3}He), ZDC (n), B0 EMCal (de-excitation γ); Z-identification if available (open question #19).
Observable	cos 2φ′ coefficient of the ratio of spin-sorted $N_f(\varphi')$ in (x, Q²) bins, $\varphi' = \varphi_e - \varphi_S$	cos 2α, cos 2β and cos(α+β) coefficients of the same ratio in (x_P, t, Q²) bins, $\alpha = \varphi_e - \varphi_S$, $\beta = \varphi_t - \varphi_S$

2 · Angles and frames

For an unpolarized electron on a spin-1 target whose alignment axis has polar angle θ_S and azimuth φ relative to the virtual photon and the lepton plane, the Hoodbhoy–Jaffe–Manohar cross section for the projection state m reads [1,2]

$$\frac{d\sigma^{(m)}}{dx\,dy\,d\phi}=\frac{2y\alpha^2}{Q^2}\left[F_1+\tfrac{2}{3}a_m b_1+\frac{1-y}{xy^2}\left(F_2+\tfrac{2}{3}a_m b_2\right)-\frac{1-y}{y^2}\,c_m\sin^2\theta_S\,\Delta(x,Q^2)\cos 2\phi\right],\qquad c_m=3m^2-2$$

and a fill with populations p_m along a transverse axis (θ_S = 90°) has the φ′ distribution 1 + a_2 cos 2φ′ with

$$a_2=-P_{zz}\frac{1-y}{y^2}\frac{\Delta}{D_\phi},\qquad D_\phi=F_1+\frac{1-y}{xy^2}F_2,\qquad \hat{A}\equiv\frac{a_2}{P_{zz}},\qquad P_{zz}=\sum_m p_m c_m.$$

Which angle. The φ of the master formula is the azimuth of the transverse spin about q from the lepton plane, in a frame in which P and q are collinear. The covariant definition used for every transverse-spin azimuth in SIDIS and exclusive analyses [3,4] applies verbatim to the spin four-vector S, and with P′ in place of S to the recoil azimuth of §4:

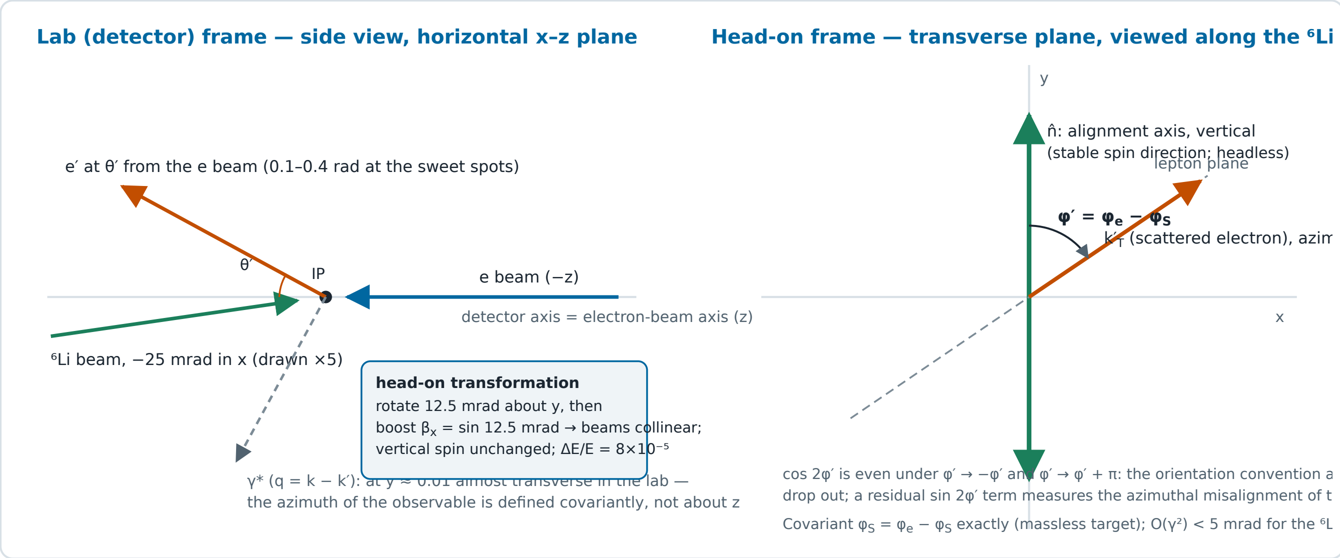
$$\cos\phi_S=-\frac{l_\perp\cdot S_\perp}{|l_\perp||S_\perp|},\qquad \sin\phi_S=-\frac{\varepsilon_{\mu\nu\rho\sigma}l^\mu S^\nu P^\rho q^\sigma}{(P\cdot q)\sqrt{1+\gamma^2}\,|l_\perp||S_\perp|},\qquad v_\perp^\mu=g_\perp^{\mu\nu}v_\nu,\quad \gamma^2=\frac{M^2Q^2}{(P\cdot q)^2}$$

with $g_\perp$ the projector orthogonal to P and q [3]. In the head-on frame — ion along +z, electron along −z, the frame of the whole fast simulation — with the spin axis transverse to the beam at lab azimuth φ_S, this reduces *exactly* to $\varphi' = \varphi_e - \varphi_S$ for a massless target and up to O(γ^2) for the nucleus: below 5 mrad for $\gamma^2 \le 0.02$, i.e. everywhere in the sweet spots (`reco.azimuth_wrt_lepton_plane`, verified against an explicit collinear-frame construction to 10^{-13}). The chain uses the covariant formula throughout, since the coherent channel needs it.

Which frame. The beams cross at 25 mrad in the horizontal plane at IP6; the ePIC axis is the electron-beam axis and the ion beam enters at −25 mrad in x. The head-on frame is a rotation of 12.5 mrad about y and a boost β_x = sin 12.5 mrad: energies change by 8×10^{-5}, the vertical spin direction not at all. For the electron the two cancel at first order, leaving an odd-harmonic azimuthal distortion of 0.7×10^{-3} (θ′ = 107 mrad) to 2.4×10^{-3} (383 mrad) in cos φ, cos 2φ untouched to 10^{-17} and the response of a true cos 2φ term one within 10^{-5} — no fake cos 2φ′ for the electron. The recoil is another matter (§4).

Orientation and the axis. cos 2φ′ is even under φ′ → −φ′ and φ′ → φ′ + π, and the axis is headless (m = +1 and m = −1 enter through $c_m = 3m^2 - 2$ symmetrically), so the orientation convention and the sign of n̂ drop out; only the direction modulo π matters. A misalignment δ turns the modulation into cos 2δ cos 2φ′ + sin 2δ sin 2φ′, and since cos 2φ′ is the only physics harmonic in the tensor ratio (m-symmetric fills set a_1 = 0; the b_1-type constant is φ′-independent), a fitted sin 2φ′ term measures δ from the data themselves. A polar tilt ε out of the transverse plane dilutes by $\sin^2\theta_S = 1 - \varepsilon^2$ (0.8% at 5°).

Harmonics at finite γ. The exact spin-1 decomposition (Cosyn et al. [12], Eq. 10) adds tensor pieces with the same azimuthal signatures: b_2, b_3, b_4 × cos φ_{TL} at order γ (Eq. 17d), b_2, b_3 × cos 2φ_{TT} at order γ^2 (Eq. 17e), γ = 2Mx/Q. At the sweet spots $\gamma^2 \le 0.023$, and the leakage into cos 2φ′ is bounded by $\gamma^2(b_1/F_1) \times 1.15$ — the full Eq. (17e) combination, ≈ 6.9× the $\gamma^2(b_1/F_1)/6$ of the b_1 term alone (plans/08 D2, correcting code review G2) — so with HERMES' $b_1 \approx 0.11$ at x = 0.012, i.e. $b_1/F_1 \approx$ 0.01–0.02, it is ≤ 0.15% of everything published here, and computable in any case from the same data's A_{zz}; the O(γ) cos φ′ term is orthogonal to the fit at full coverage. For the axis along q the T_{LT} and T_{TT} parameters vanish (Eq. 20), which is why the transverse axis is the only choice for Δ. At x ≳ 0.2 and $Q^2 \approx 1$ GeV² ($\gamma^2 \approx 0.1$) the leakage exceeds a $\Delta/F_1 \sim 10^{-3}$ scenario outright and the A_{zz} subtraction is required (code review G2).



Schematic 1 — frames and the azimuth of the observable. Left: the detector frame with the 25 mrad crossing angle, the electron measured at (E′, θ′, φ_e) about the electron-beam axis, the virtual photon far from that axis at low y. Right: in the head-on frame the alignment axis is vertical and the physics azimuth is the angle between the lepton plane and the axis, which the covariant formula reproduces as $\varphi_e - \varphi_S$.

3 · The inclusive chain

1. **Event selection and electron identification.** One e′ candidate, track matched to an EMCal cluster; E/p and shower-shape cuts against photoproduced π^-, and a charge-symmetric subtraction for the pair-symmetric background at high y. $\varepsilon_{\mathrm{eID}}(\eta)$ = 0.70–0.95 on the ATHENA/ECCE anchors of the money_delta study; no official ePIC curve exists.
2. **Scattered-electron kinematics.** E′ from the calorimeter (2%/√E ⊕ 1%, 1.2% at E′ ≈ E_e) or the tracker (3% at η ≈ −3 in the tracking-only table), θ′ and φ_e from the track at the vertex (1–3 mrad), transformed to the head-on frame; k, k′, P and S = (0, n̂) enter the covariant azimuth.
3. **Kinematic reconstruction.** The electron method,

$$Q_e^2 = 4E_eE'\sin^2\frac{\theta'}{2}, \qquad y_e = 1-\frac{E'}{E_e}\cos^2\frac{\theta'}{2}, \qquad x_e = \frac{Q_e^2}{s\,y_e}, \qquad \frac{\delta y_e}{y_e} = \frac{1-y}{y}\left[\frac{\delta E'}{E'} \oplus \tan\frac{\theta'}{2}\,\delta\theta' \oplus \frac{\delta E_e}{E_e}\right]$$

has a good Q² (δQ²/Q² ≈ δE′/E′ ⊕ cot(θ′/2)δθ′, angle-dominated at the sweet spots: 5.2% for the 3 mrad track table of §7 at θ′ ≈ 0.1 rad, 1.6% at spot 4, 2% if the backward disks reach 1 mrad — that table is an unsourced placeholder) but loses y as (1 − y)/y: at y = 0.01 a 1.2% energy resolution gives δy/y = 120%, the electron-ring energy spread alone (≈ 10⁻³) 10%. Table 1 places money plot 5: at the mid energy three of the four spots are at y = 0.010–0.026, the electron carrying 98–99% of the beam energy at η ≈ −2.9 to −2.4, and at the top energy all four sit at y ≈ 0.01. The remedy is the one HERA used at low y [5,6]: y from the hadronic final state,

$$y_{JB} = \frac{\Sigma}{2E_e}, \qquad y_\Sigma = \frac{\Sigma}{\Sigma + E'(1-\cos\theta_e)}, \qquad \Sigma = \sum_{h\neq e'} (E_h - p_{z,h}), \qquad x = \frac{Q_e^2}{s\,y_\Sigma}\ \text{(mixed method)}$$

with θ_e measured from the ion direction. The Σ method is insensitive to collinear initial-state radiation and both methods to particles lost along the ion beam (E − p_z ≈ 0 for forward remnants) — but the insensitivity is to *y* alone: the mixed x = $Q^2_e/(s\,y_\Sigma)$ takes its numerator from the electron, so a photon radiated before the vertex carries x → x/(1 − z) exactly, and the variable the analysis bins in is not ISR-robust even though the y it is built from is (plans/07 WP4); their resolution is set by hadronic calorimetry and by losses in the electron direction. Three documented expectations bracket it: ATHENA, e−Σ or DA for y ≲ 0.1, δy/y ≈ 25% at y ≈ 0.01 and ≈ 10% at y ≈ 0.1, JB 20–30% throughout (Sec. 3.1, Fig. 22) [18]; the ePIC kinematic-fit study, hadronic E − p_z smeared by $\sigma(\delta_h)/\delta_h$ = 25%, Δy/y widths 0.2–0.3 in 0.01 < y < 0.05 [19]; the ATHENA fast simulation of Arratia et al., RMS(y)/y ≈ 0.13–0.17 for IΣ, DA and JB at y ≈ 0.05–0.2 [20]. The band assumed here is 15–30%, 25% central (the value used in §7). The low-energy configuration moves the same (x, Q²) to y = 0.05–0.25 (Table 1), so the energy scan is part of the reconstruction strategy for the x ≈ 0.1 bins.

SWEET SPOT	X, Q² [GEV²]	E(5)×⁶LI(40.8): Y	E(10)×⁶LI(99.5): Y, E′ [GEV], Θ′ [MRAD], Η	E(18)×⁶LI(137.5): Y	ΔY/Y AT MID, E′ ALONE (1.2%)
1	0.056, 1.14	0.025	0.0051, 9.98, 107, −2.93	0.0020	2.3
2	0.022, 1.14	0.062	0.0128, 9.90, 107, −2.92	0.0051	0.93
3	0.141, 3.13	0.027	0.0056, 10.02, 177, −2.42	0.0022	2.1
4	0.141, 14.3	0.124	0.0255, 10.10, 378, −1.65	0.0102	0.46

Table 1 — the four sweet-spot super-bins of money plot 5, and the same (x, Q^2) at all three energies ($y = Q^2/(sx)$). **Recomputed 2026-08-27** at the corrected beam energies: EIC ions are γ-matched rather than rigidity-scaled, so ^{6}Li sits at 40.8 / 99.5 / 137.5 GeV/u and not 20.5 / 50 / 137.5 (plans/10). Every y halves, and the electron method degrades with it — δy/y at the mid configuration goes from 1.2 / 0.46 / 1.1 / 0.23 to 2.3 / 0.93 / 2.1 / 0.46, which *strengthens* rather than weakens the finding below. E′, θ′ and η are unchanged to the quoted precision: at $y \ll 1$ the scattered electron barely knows the ion energy. Figure 1(b) places all three energies.

The hadronic final state in the chain. `polligen.hfs` carries the hadronic final state of unpolarized DIS events through a hadron-side response and the Σ, Jacquet–Blondel and double-angle methods, as a library for the pseudo-experiments: the captured fraction of $\Sigma = \Sigma(E - p_z)$ and the p_T ratio of a generator event from the same (x, Q^2) cell are applied to the pseudo-event's exact truth, noise is added per event, and y follows with the *reconstructed* electron. The tensor polarization enters only the azimuthal weight, so an unpolarized generator is the right source: PYTHIA 8, 8 M events over the three beam configurations, e + p and e + n at the per-nucleon energies with lepton radiation off (`tools/pythia8/gen_dis_hfs.py` ; manifest in `evgen/samples/README.md`), with `hfs.ToyHFS` as a labelled fallback. The response: tracks within $|\eta| \le 3.5$ above $p_T = 0.2$ GeV (95% plateau), calorimeter objects within $|\eta| \le 3.7$ (EMCal photons above 0.1 GeV, neutral and untracked charged hadrons in the HCal above 0.5 GeV at 90%), Yellow Report resolutions per region, and Gaussian noise on Σ and on each p_T component — 50 MeV by default, the magnitude Arratia et al. needed to reproduce the H1 full simulation at low y [20]. Figure 4 and Table 1b: 78–91% of Σ is within the acceptance and above threshold at the four sweet spots (tracks 0.44–0.55, photons 0.25–0.27, HCal objects 0.09–0.10), the rest escaping forward beyond $|\eta| = 3.7$ (7–19%) or lying below threshold (1–6%), and 69–85% is captured through the full response; without noise the Σ method resolves y to 19–22% throughout $0.005 < y < 0.05$, and with 50 MeV of noise the four sweet spots give 0.32 / 0.22 / 0.29 / 0.15 (Σ and JB alike; DA 0.51 / 0.40 / 0.45 / 0.22; the electron method 1.18 / 0.46 / 1.06 / 0.46). The noise floor, not the fragmentation, decides the lowest bins: at y = 0.005 the Σ resolution runs 0.22 / 0.33 / 0.54 / 1.01 for 0 / 25 / 50 / 100 MeV of noise and at y = 0.0125 0.19 / 0.21 / 0.28 / 0.44, so the ePIC noise and threshold floor at $\Sigma_h \approx 0.2$–0.5 GeV sets the $y \approx 0.01$ bins (plans/04 #21). Each configuration selects its own sweet spots, which move to lower y as s grows: they resolve to 0.38 / 0.23 / 0.24 / 0.11 at the low configuration and 0.23 / 0.19 / 0.21 / 0.18 at the top, so the $x \approx 0.14$ bins are best measured at the low-energy configuration (0.24 at $Q^2 = 3.1$ and 0.11 at 14 GeV²) (`hfs_resolution.py --config 0/1/2`).

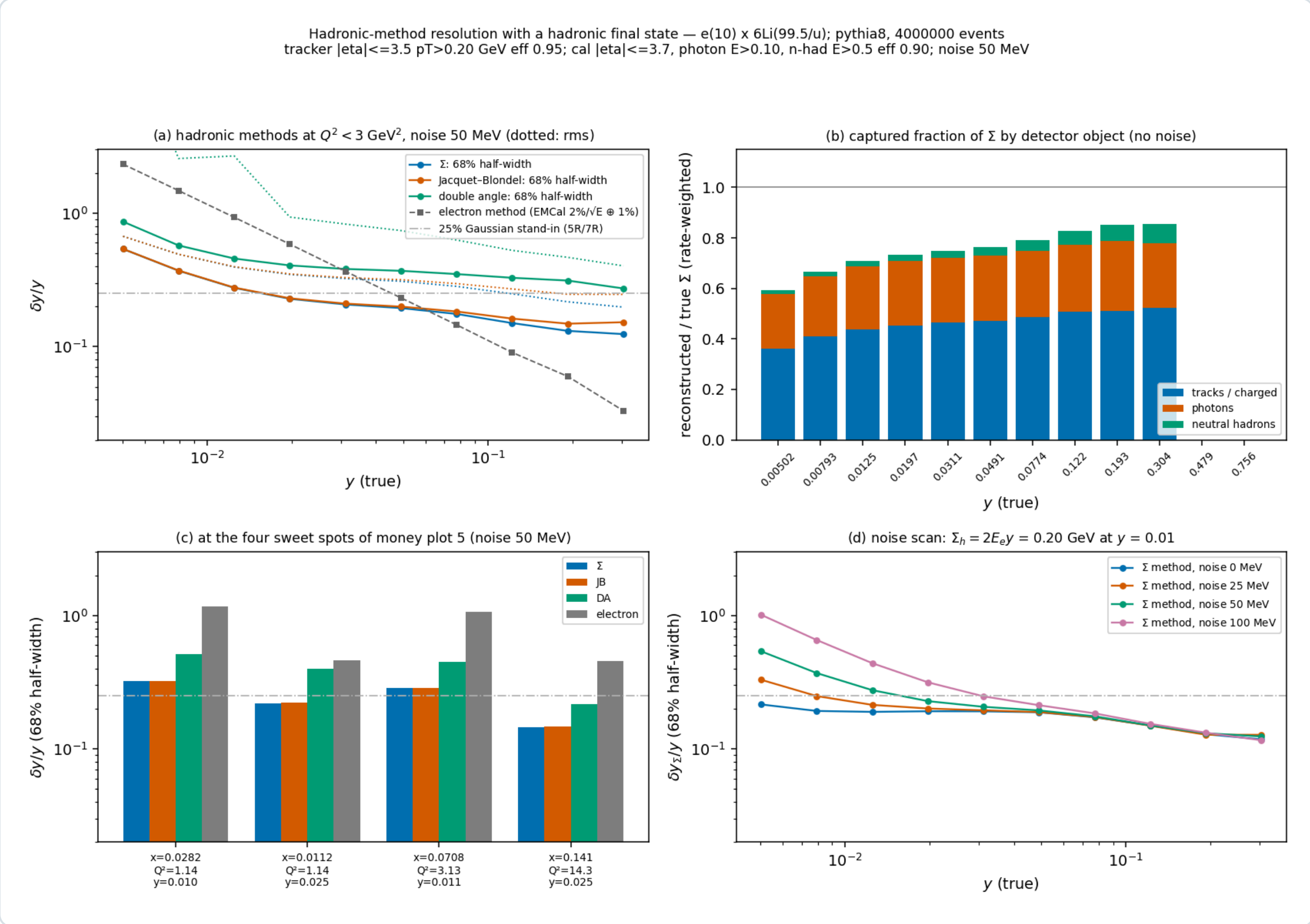


Figure 3 — hadronic-method resolution with a hadronic final state (e10 × ⁶Li99.5, **PYTHIA 8**, 4 M e + p and e + n events, `hfs_resolution.py`). **(a)** δy/y (68% half-width; dotted: rms) of the Σ, Jacquet–Blondel and double-angle methods at $Q^2 < 3$ GeV² with the default response and 50 MeV noise, against the electron method and the 25% stand-in. **(b)** Captured fraction of Σ by particle species, rate-weighted at $Q^2 < 3$ GeV² and through the full response, no noise (0.66–0.75 across the sweet-spot y range; Table 1b is the Σ-weighted decomposition at the spots themselves). **(c)** The four sweet spots. **(d)** The Σ-method resolution for 0, 25, 50 and 100 MeV of calorimeter noise.

Where the hadronic $E - p_z$ sum goes. Each particle contributes $E - p_z = m_T e^{-y}$ to Σ_h ($p_T e^{-\eta}$ for a massless one), an exponentially falling share of what goes forward — except the mass term of forward baryons, which is what the target remnant carries — so whether the hadronic final state is "in acceptance" is a question about where Σ is carried, not about where the particles are. Figure 4 and Table 1b answer it at the four sweet spots on the PYTHIA 8 sample (`hfs_acceptance.py`). The struck-quark direction, the double-angle hadronic angle γ_h, sits at η = 0.7–2.1, inside the tracker; but Σ is carried by particles whose Σ-weighted median η is 1.7–2.6 and, at the y ≈ 0.01 spots — a low-mass system, W ≈ 6 GeV with five charged particles — whose 84th percentile reaches the calorimeter edge at 3.7–3.9. Geometry and thresholds alone leave 78–91% of Σ within the acceptance and above threshold: 16–19% escapes forward beyond |η| = 3.7 at y ≈ 0.01 (7–8% at the y ≈ 0.025 spots), 60% of it remnant nucleons whose $E - p_z$ is the mass term, thresholds cost 1–6%, and nothing is lost backward. Through the full response — efficiencies, the track turn-on, the pion-mass hypothesis on tracks and massless clusters — the captured fraction is 0.69 / 0.74 / 0.73 / 0.85 (Σ-weighted; per-event median 0.72 / 0.78 / 0.76 / 0.87), and the median y_Σ/y is the same 0.72–0.87. The far-forward lithium fragments do not enter: a beam-rapidity α, deuteron or intact recoil carries $E - p_z = m^2/(2p)$ — 4.4 MeV per nucleon, 17 MeV for an α, 26 MeV for an intact ⁶Li — a per-event constant at η ≈ 8 that never reaches the measured Σ_h whether it is tagged or not, which is why this method is independent of the tagging question. Two consequences follow. The loss is a *scale bias* on y_Σ, 13–28% through the response, and the response's (x, Q²) library reproduces it in the pseudo-experiments — each pseudo-event takes the captured fraction of a library event from its cell — rather than correcting it; the migration-inclusive bin-centering factor K of §7 absorbs it to the extent the response is right, and an analysis would calibrate it from the same simulation, which `HFSResponse(calibrate=True)` now does with the library's per-cell mean (§7 gives the calibrated 5R numbers, and a residual 1% scale error moves $\mathcal{A}$ by 0.2–0.7%). And the loss is not what sets the resolution. Extending the calorimeters to the ePIC nominal |η| = 4.0 recovers about a quarter of the forward loss (19 → 14% at the first spot) and moves the median y_Σ/y from 0.72 to 0.74, but leaves δy/y unchanged: the forward share is a nearly constant fraction of Σ_h event to event, so removing it entirely moves the median, not the width. What sets the width is the 50 MeV noise at the y ≈ 0.01 spots (0.25 / 0.23 of the 0.32 / 0.29) and, at the y ≈ 0.025 spots where Σ_h = 0.5 GeV, the within-acceptance fluctuation of the captured share — thresholds, efficiencies and the massless treatment of hadrons (0.19 of the 0.22 at zero noise at the second spot, comparable to the noise at the fourth). The B0 region (η ≈ 4.6–5.9) is not in the response and would catch part of the remainder.

SWEET SPOT (X, Q²)	Y	W [GEV]	⟨N_{CH}⟩	Γ_H AS H · Σ-WEIGHTED PARTICLE H (16 / 50 / 84%)	WITHIN ACCEPTANCE AND ABOVE THRESHOLD, |H| ≤ 3.7 (TRACKS / PHOTONS / HCAL)	LOST FORWARD	BELOW THRESHOLD	SAME, |H| ≤ 4.0	CAPTURED THROUGH THE RESPONSE, 3.7 (MEAN · MEDIAN)	ΔY/Y (Σ), 3.7 / 4.0
0.028, 1.14	0.010	6.3	4.6	1.6 · 1.8 / 2.6 / 3.9	0.78 (0.44 / 0.25 / 0.10)	0.19	0.03	0.83	0.69 · 0.72	0.32 / 0.32
0.011, 1.14	0.025	10.0	5.9	0.7 · 0.8 / 1.7 / 2.9	0.86 (0.51 / 0.26 / 0.10)	0.08	0.06	0.88	0.74 · 0.78	0.22 / 0.22
0.071, 3.13	0.011	6.5	4.6	2.1 · 1.9 / 2.6 / 3.7	0.82 (0.47 / 0.25 / 0.09)	0.16	0.02	0.86	0.73 · 0.76	0.29 / 0.28
0.141, 14.3	0.025	9.4	5.6	2.0 · 1.7 / 2.2 / 2.9	0.91 (0.55 / 0.27 / 0.09)	0.07	0.01	0.93	0.85 · 0.87	0.14 / 0.14

Table 1b — the true Σ_h at the four sweet spots of money plot 5 (mid configuration, PYTHIA 8, e + p and e + n merged) by fate. The acceptance columns are geometry and thresholds only — tracker |η| ≤ 3.5, p_T > 0.2 GeV; photons > 0.1 GeV, HCal objects > 0.5 GeV, "HCal" including charged particles the tracker does not see; no efficiencies, smearing or mass hypotheses — for the response's calorimeter reach and for the ePIC nominal 4.0; "lost backward" is below 0.1% everywhere. The "captured through the response" column applies the full `HadronResponse` (95% / 90% efficiencies, the track turn-on, the pion-mass hypothesis on tracks and massless clusters; noise off) and equals the median y_Σ/y. The Σ-method δy/y is through the full response with 50 MeV of noise; the electron method gives 1.2 / 0.46 / 1.1 / 0.46. Differences of 0.01 between this table, Figure 3 and the two calorimeter reaches are Monte-Carlo noise.

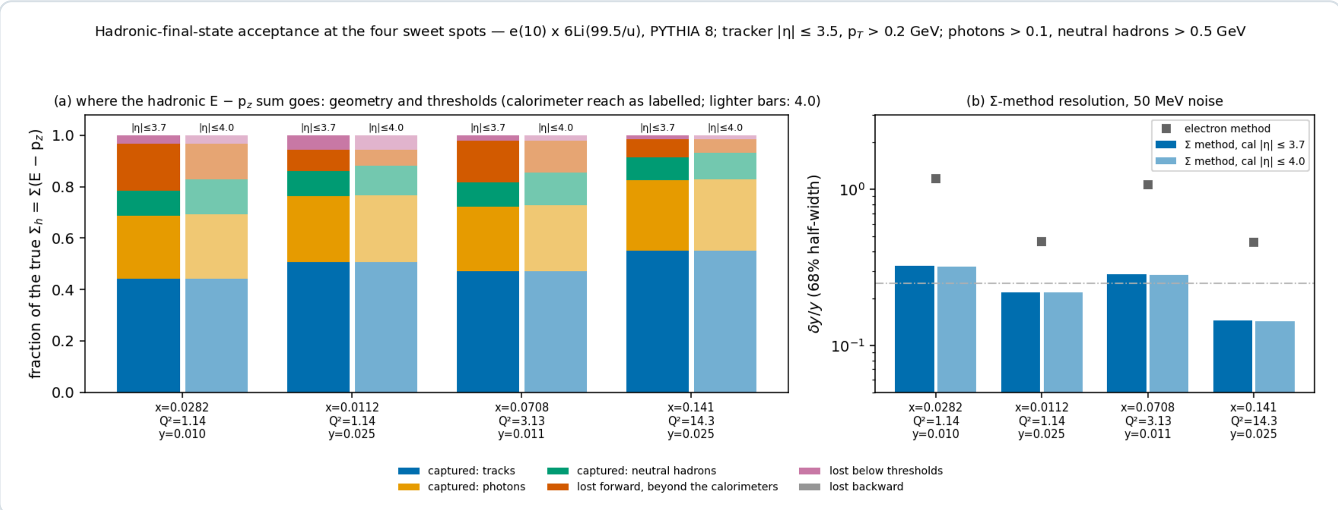


Figure 4 — hadronic-final-state acceptance at the four sweet spots (e10 × ⁶Li99.5, PYTHIA 8, `hfs_acceptance.py`). **(a)** The true Σ_h by fate, geometry and thresholds only (no efficiencies) — within acceptance as tracks, photons and HCal objects; lost forward beyond the calorimeters, below thresholds, or backward — for a calorimeter reach of |η| ≤ 3.7 (full bars) and 4.0 (lighter). **(b)** The Σ-method δy/y through the full response with 50 MeV of noise for both reaches, against the electron method: the extra reach moves the bias, not the resolution.

4. **The azimuth.** φ′ = $\varphi_e - \varphi_S$ from the covariant formula (§2), φ_S being the alignment-axis azimuth of that bunch. The track direction resolution enters as $\sigma_\varphi = \sigma_\theta/\sin\theta'$ ≈ 20–30 mrad at the sweet spots and dilutes cos 2φ′ by ⟨cos 2δφ′⟩ =

0.985–0.999: negligible.

5. **Binning in reconstructed (x, Q²).** Figure 1(c): with $\delta y_{had}/y$ = 15% (20%, 30%) the 3×3 super-bins keep a purity of 0.81–0.83 (0.74–0.77, 0.61–0.65) and the amplitude seen in the reconstructed bin is 0.98–0.99 (0.96–0.99, 0.92–0.98) of the true-bin value, because A(x) varies slowly across the bin; with the electron method alone the low-y spots collapse to purities of 0.41–0.56 with 26–53% of the events surviving. The correction is the standard migration matrix, built on the hadronic-method resolution.
6. **Spin-state sorting and the estimator.** The events of each (x, Q²) bin are histogrammed in φ′ per spin state f, $N_f(\varphi')$. Any smooth ε(φ′) multiplies every N_f identically, so the bin-wise ratio

$$R(\phi')=\frac{\sum_f w_f N_f(\phi')}{\sum_f N_f(\phi')},\qquad w_f=P_f-\bar{P},\qquad \bar{P}=\frac{\sum_f L_f P_f}{\sum_f L_f},\qquad R=\frac{\sigma_P^2\,T(\phi')}{1+\bar{P}\,T(\phi')},\qquad T=\kappa+\hat{A}\cos 2\phi',\qquad \sigma_P^2=\frac{\sum_f L_f(P_f-\bar{P})^2}{\sum_f L_f}$$

is free of ε(φ′) exactly (raw counts with luminosity-weighted w_f, so that $\Sigma_f L_f w_f = 0$), and a relative-luminosity error enters only through P̄, as a φ′-independent constant plus a second-order rescaling (δ × P̄/σ_P^2). It presumes both spin states see the *same* ε(φ′) — the one systematic the ratio cannot cancel, and the reason the states must alternate bunch by bunch within a fill or the harmonic be stable to 10^{-4} between fills (§7.1). A weighted least-squares fit of T(φ′) = c + d cos 2φ′ (+ e sin 2φ′), T obtained from R bin by bin by an exact inversion, gives Â = d once the finite-bin dilution sin w/w is divided out; κ = c is the transverse tensor asymmetry of the b_1 sector, a by-product whose sign follows the program's b_1 transcription rather than the HJM/HERMES convention (Cosyn et al. [12], Eq. 27: $A_{zz} = -(2/3)b_1/F_1$; code review G1) and changes nothing in the Δ sector. For two fill types of equal luminosity the statistical error is

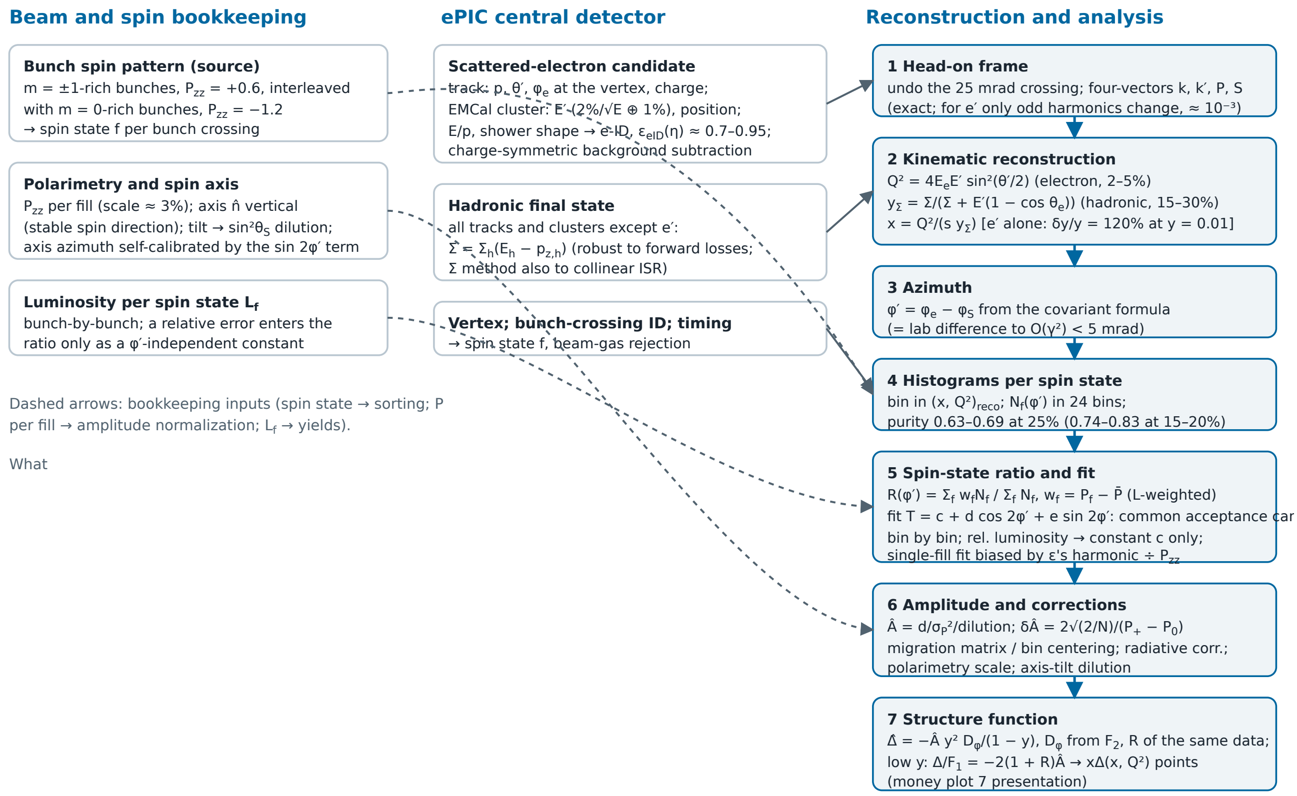
$$\delta\hat{A}=\frac{2}{P_+-P_0}\sqrt{\frac{2}{N}}\ \text{(two fills)}\qquad \text{vs}\qquad \delta\hat{A}=\frac{1}{P_{zz}}\sqrt{\frac{2}{N}}\ \text{(single fill)}$$

so m = ±1-rich bunches at P_+ = +0.6 alternated with m = 0-rich bunches at P_0 = −1.2 (the same source purity gives $-2P_+$, the pattern of the A_{zz} run plan) are 1.5× more precise than a single P_{zz} = 0.6 fill with the whole luminosity; an unpolarized reference fill (P_0 = 0) would cost a factor two instead. Figure 1(d): under ε = 1 + 0.03 cos 2φ′ + 0.02 cos φ′ the single-fill fit returns Â = 0.062 for a true 0.0118 — the acceptance harmonic divided by P_{zz} — while the ratio is unbiased with the analytic error. A percent-level cos 2φ harmonic aligned with the vertical axis is to be expected: the crossing-angle plane is horizontal, module boundaries and dead areas are not azimuthally random, and the Roman-Pot cutout is a rectangle.

7. **Amplitude to Δ.** With F_2 and R = σ_L/σ_T of ⁶Li per nucleon measured in the same data (the unpolarized cross section of the same sample, with its own radiative corrections) or taken from nuclear-PDF fits,

$$\hat{\Delta}(x,Q^2)=-\hat{A}\frac{y^2 D_\phi}{1-y}\ \rightarrow\ -\hat{A}\frac{F_2}{x}=-2(1+R)F_1\hat{A}\quad (y\ll 1),\qquad \text{i.e.}\ \ \frac{\Delta}{F_1}=-2(1+R)\hat{A}$$

At the sweet spots the conversion is insensitive to y — the (1 − y)/y² factor that makes low y attractive for the amplitude cancels against D_φ — so Δ/F_1 is essentially a direct ratio measurement, and the y mis-measurement of the electron method moves the x label of the point, not the amplitude. Of the two structure-function inputs only R moves the answer: F_2 cancels *exactly* in the cos 2φ amplitude (2×10^{-16} residual, two independent levers). The chain *can* take R from the published SLAC/E143 R1998 world fit [21], threaded behind the `r_func` hook, but its default — and every number in this note — is still the simplified placeholder; switching would move Δ/F_1 by +17.8 / +18.5 / +7.6 / −4.4% at the four sweet spots. Remaining corrections: the tensor rate shift $(2/3)a_m(b_1 + \ldots)$ in D_φ, at the 10^{-3} level and measured by κ; target-mass terms of order γ²; radiative corrections, uncalculated for tensor observables (plans/07 WP4); the per-nucleon and dilution conventions, which are interpretation rather than reconstruction; and the polarimetry scale $\delta P_{zz}/P_{zz}$ ≈ 3%, common with A_{zz}.



`polligen.reco` + `.hfs` + `.recopseudo` implement: exact frames and angles, Gaussian $E'/\theta'/\varphi e$ smearing, η-dependent e-ID, a hadronic final state through the hadron-side response, a smooth $\varepsilon(\varphi')$, spin-sorted Poisson histograms, the ratio fit (§7). What stays for Phase 2: e-ID purity, radiative events, the real φ acceptance map and its fill-to-fill stability, beam-gas backgrounds.

Schematic 2 — the inclusive reconstruction chain. Left and middle: what is recorded; right: the seven reconstruction and analysis steps of §3 with the quantity each produces and its dominant limitation. The observable is the cos $2\varphi'$ coefficient of the spin-state ratio (step 5), not the modulation of a single histogram.

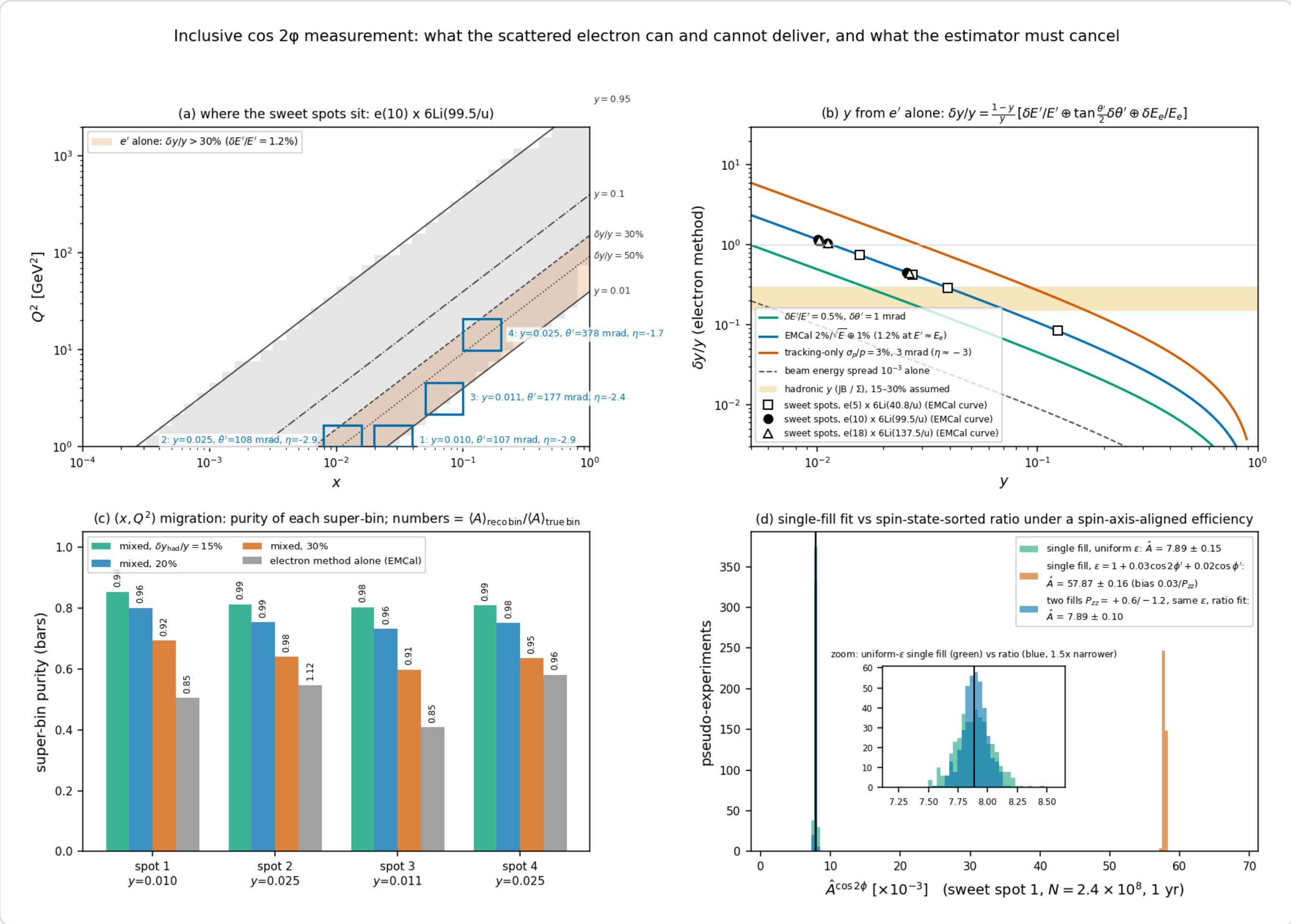


Figure 1 — the inclusive measurement quantified (e10 × ⁶Li99.5, `reco_chain_figures.py`). **(a)** The analysis acceptance with iso-y lines and the four sweet-spot super-bins; the shaded band is where the electron alone gives δy/y > 30% at a 1.2% energy resolution — spots 1–3 inside it, spot 4 just above. **(b)** The electron-method δy/y versus y for calorimeter, tracking and beam-energy-spread inputs, with the sweet spots of the three energies and the assumed 15–30% hadronic band. **(c)** Purity of each super-bin for three hadronic-y resolutions and for the electron method alone; the numbers give the amplitude averaged over the events landing in the reconstructed bin over that of the true bin. **(d)** Four hundred pseudo-experiments of the spot-1 super-bin at 1-year statistics: single-fill fit with a uniform efficiency (green), the same under ε = 1 + 0.03 cos 2φ′ + 0.02 cos φ′ (vermillion, biased by $0.03/P_{zz}$) and the two-fill ratio fit under that efficiency (blue, unbiased and 1.5× more precise; inset).

4 · The coherent chain

The coherent measurement keeps steps 1–7 for the electron and adds the recoil. Two facts fix its kinematics. The recoil keeps the beam rigidity (R = 1.000 for A/Z = 2, and $x_P < 0.01$ is below the Roman-Pot momentum resolution), so the only measured recoil quantity is its transverse displacement from the beam orbit; and coherence confines the sample to $x \lesssim 0.01$, i.e. $y \gtrsim 0.05$ at the mid energy (median $x = 3.5\times10^{-3}$, $y \approx 0.2$), where the low-y problem of §3 does not arise.

1. **Roman-Pot track.** Hits in the two AC-LGAD stations (≈ 30 ps timing for bunch association) form a local track, which the optics transfer matrices referenced to the beam orbit turn into the IP angle pair (θ_x, θ_y) relative to the ion axis; crossing angle and reference orbit drop out by construction, the central-detector vertex supplying the position term. x_L is unresolved and set to 1, so $p_T = A\, p_u\sqrt{\theta_x^2 + \theta_y^2}$ and $\varphi_t = \mathrm{atan2}(\theta_y, \theta_x)$. The bunch's own divergence dominates the resolution: $\delta p_T = A\, p_u \sigma_\theta = 22$ MeV at 99.5 GeV/u for $\sigma_\theta = 73$ μrad, hence $\sigma(\varphi_t) = 0.09$ rad at $p_T = 0.25$ GeV and a cos $2\varphi_t$ dilution of 1.5% (Figure 2d). The ePIC far-forward simulation with all beam effects gives $\Delta p_T \approx 40$ MeV for 275 GeV protons ($\Delta p_T/p_T \approx 0.09$ at 0.45 GeV, 0.04 at 1 GeV), the detector alone ≤ 1.5% — "beam effects the dominant source of momentum smearing" [17]; ZEUS's leading-proton spectrometer was limited the same way ($\sigma(t)/t = 0.14$ GeV/√|t|, Φ resolution 0.2 rad) [13].
2. **Momentum transfer.** Exactly, with $x_L = 1 - x_P$ the plus-momentum fraction,

$$t=-\frac{p_T^2+M^2x_P^2}{1-x_P}\simeq -p_T^2,\qquad t_{\min}=\frac{M^2x_P^2}{1-x_P}=3.2\times10^{-3}\ \mathrm{GeV}^2\ (x_P=0.01),\qquad x_P=\frac{M_X^2+Q^2-t}{W^2+Q^2-M^2}$$

so the Roman Pots give t, while x_P (or $\beta = x_A/x_P$) must come from the diffractive mass M_X of the central detector (the HERA M_X method [16]) — a coarse variable, a few bins, enough for the x_P-dependence test of the two-component fit. The covariant recoil azimuth differs from $\varphi_e - \varphi_t$ by less than 0.2 mrad here.

3. **Exclusivity selection.** A beam-rigidity Roman-Pot track above the envelope; no OMD track (p at R = 0.5, ^{3}He at 0.75 in the main window [8]), no ZDC neutron, no B0 EMCal photon (the 3.56 MeV state's de-excitation γ, boosted to 0.1–0.4 GeV); a rapidity gap; Z-identification if the pots resolve Z = 3 from 2 and 1 (open question #19), otherwise a two-component fit of the |t| spectrum against the ten-times-flatter α + d breakup slope; timing and vertex association against beam-gas coincidences. The incoherent cross section is m-state-blind, so residual breakup dilutes the tensor ratio without faking it, and its fraction comes from the same |t| fit.
4. **Three azimuths, two harmonics.** With the recoil measured there are two physical azimuths besides the spin: $\alpha = \varphi_e - \varphi_S$ and $\beta = \varphi_t - \varphi_S$, with $\alpha - \beta = \varphi_e - \varphi_t$ the lepton-plane–recoil angle of unpolarized diffractive DIS. The rank-2 polarization enters only through $e^{\pm 2i\varphi_S}$, so at leading power in the momentum transfer the spin-sorted distribution is

$$N_f(\alpha,\beta)\propto\varepsilon(\alpha,\beta)[1+u_1\cos(\alpha-\beta)+u_2\cos 2(\alpha-\beta)+P_f(\kappa+a_e\cos 2\alpha+a_t\cos 2\beta+a_m\cos(\alpha+\beta))]$$

The deformation term of the polarized-deuteron anchor — the diffractive slope modulated by the aligned gluon density, $a_2(t) \propto |t|$, digitized in `coherent.MANTYSAARI_A2_DEUTERON` — is a_t: it lives in the *recoil* azimuth β, needs no photon helicity flip, and vanishes as t → 0. (The anchor's Φ is defined in the γ*d c.m. frame, where at $Q^2 > 0$ the transformation from the lab "would mix the polarization states" [9] — the O(γ) mixing the covariant definitions of §2 absorb.) The double-helicity-flip term, the coherent counterpart of Δ, needs the photon's linear polarization and hence the lepton plane: it is a_e, lives in α, and is flat in t at leading power. The two mechanisms are therefore separated by the azimuth they modulate as well as by their t-dependence; a_m and the unpolarized $u_{1,2}$ are nuisances. The latter are the L–T and T–T′ interference terms of diffractive DIS with a detected recoil, $u_1 = (2-y)\sqrt{(1-y)}\,A_{LT}/[2(1-y)+y^2]$ and $u_2 = 2(1-y)\,A_{TT}/[2(1-y)+y^2]$ [14]; ZEUS measured $A_{LT} = -0.036 \pm 0.036$, $A_{TT} = -0.030 \pm 0.037$ for $x_P < 0.01$ and $+0.051 \pm 0.024$, -0.010 ± 0.024 for $0.01 < x_P < 0.1$ [13], and the pQCD LT-interference model gives $A_{LT} \approx 0.03$ at our $\beta \approx 0.5$, $p_T/Q \approx 0.2$ [15]; §7 uses $u_1 = 0.05$, $u_2 = 0.02$, at the edge of those bounds. The measurement histograms (α, β) in two dimensions per (x_P, t, Q^2) bin and fits a_t and a_e from the spin-state ratio R(α, β) — money plot 6 places the anchored $\langle a_2\rangle$ on the electron axis α, harmless for a statistical projection — with u(α − β) carried, since it survives in the ratio's denominator. An *error* δu does not stay second order: it reaches the pure harmonics as $a_t\cdot\delta u_2\cdot\langle\cos 2\beta\rangle$ and $a_e\cdot\delta u_1$, and with $|\langle\cos 2\beta\rangle| \approx 0.77$ from the cutout and $a_t/a_e \approx$ 12–34 the first dominates: a ZEUS-1σ $\delta u_2 = 0.024$ shifts a_e by 20% in the lowest |t| bin — 1.6 one-year statistical errors — rising to 35% in the highest, while the same δu_1 moves it by under 1%. u_2, not u_1, is the nuisance to measure in situ from the spin-averaged counts.

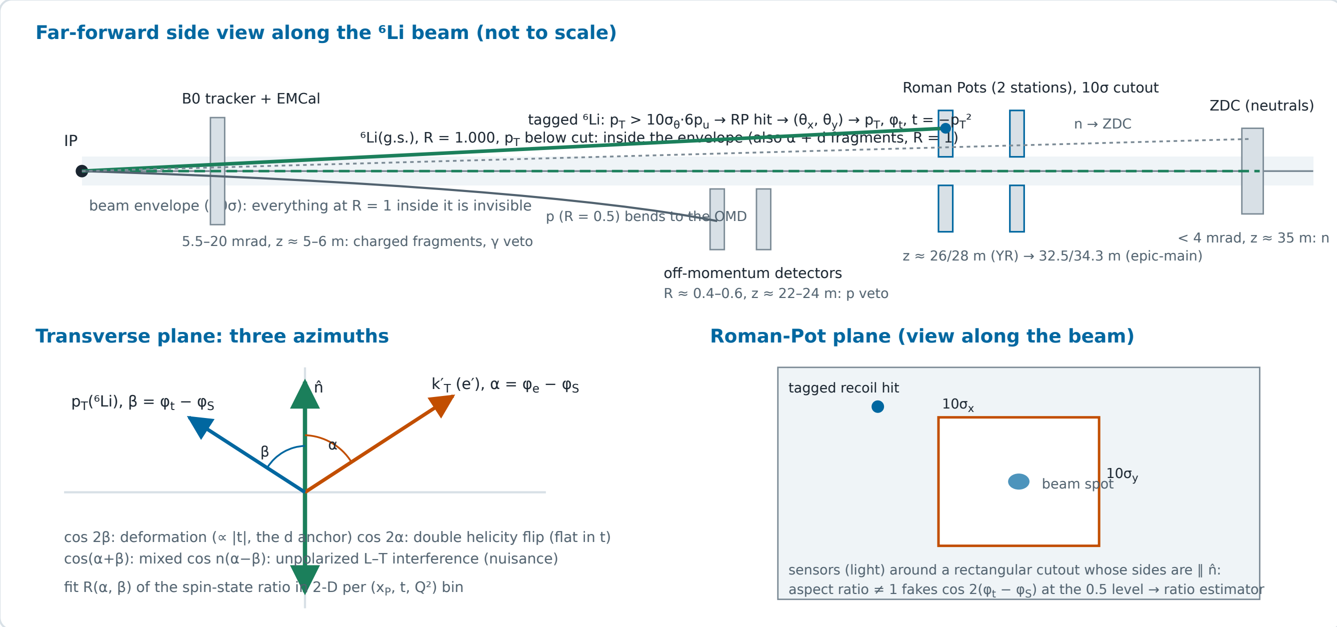
The forbidden harmonics. With unpolarized electrons and a headless axis the cross section is even under (α, β) → (−α, −β), so sin 2α, sin 2β and sin(α+β) vanish exactly: the coherent null test, counterpart of the inclusive sin 2φ′ term. Their pattern says *which* azimuth is misaligned — a spin-axis error δ gives sin/cos = tan 2δ in α *and* β, a Roman-Pot azimuthal roll δ_t gives tan $2\delta_t$ in β alone and leaves the lepton-plane harmonic exactly null (verified to 10^{-5} in `tests/test_recopseudo.py`). The recoil null is the sensitive one, resolving a pot roll in a year to 5–8 mrad in the three |t| bins that carry the statistics and 16 mrad in the highest, against the α null's 60–200 mrad on the spin axis and 2.3 rad in that highest bin. The sin columns are orthogonal to the cos ones on the full grid, so the null costs nothing; its t-dependent sine must use the same t-shape template as its cosine partner, since with a plain ⟨sin 2β⟩ it misses its own closure by 5%.

5. **Acceptance geometry and the estimator.** The acceptance is not a circular p_T cut: the pots are 25.6 × 12.8 cm AC-LGAD planes surrounding a *horizontal slot* through which the beam, its momentum spread and its dispersion pass [17], and its symmetry axes are horizontal and vertical — parallel and perpendicular to the spin axis. Anisotropy is the norm: HERA's proton beam had a transverse-momentum spread at the IP of 45 MeV horizontally against 100 MeV

vertically [13]. For the exponential recoil spectrum the tagged azimuthal distribution is $\varepsilon(\varphi_t) = \exp[-B\ \rho(\varphi_t)^2]$ with ρ the cutout boundary, and its $\langle\cos 2\varphi_t\rangle$ about the horizontal runs from 0 for a square through +0.49 at aspect ratio c_y/c_x = 1.25 and +0.71 at 1.5 to −0.36 to −0.45 for the slot-like ratios 0.3–0.6 of Figure 2b and −0.77 for the wider slot assumed before the aperture was measured (0.55 × 0.22 GeV) — ten to twenty times the anchored $a_t \approx 0.036$, about the same symmetry axis (the sign flips when the axis is referred to the vertical spin direction). A square still carries $\langle\cos 4\varphi_t\rangle = 0.34$, which mixes into cos 2β at second order through the physics term. No single-fill fit survives this; the spin-state ratio of §3 cancels it exactly, bin by bin in (α, β).

6. **The near-beam cut is angular, and the aperture is measured.** The documented 0.20 GeV (high-acceptance) and ≈ 0.41–0.45 GeV (high-divergence) cuts are ten beam divergences times the 275 GeV proton momentum [7], $\sigma_\theta \approx 73$ and 149 μrad. The same envelope on a ^{6}Li of momentum $6p_u$ is $10\sigma_\theta \cdot 6p_u$ — 0.18, 0.43 and 0.60 GeV at 40.8, 99.5 and 137.5 GeV/u — for circular tag acceptances of 20%, 8×10^{-5} and 1.5×10^{-8} at B = 50 GeV^{-2} (Figure 2c), and 5×10^{-7} / 8×10^{-26} / 4×10^{-13} once the Yellow Report's per-configuration divergences replace the proton-derived 73 μrad (plans/10), against the projection's constant 0.20 GeV cut (13.5%). The $A\ p_u$ scaling is kinematics; the lithium divergence itself is undocumented (plans/04 #11). Against that envelope the optics scan (`coherent_optics_scan.py`) gives, at the slot cutout, tagged fractions 32% / 3.0% / 4×10^{-5} / 2×10^{-7} at 0.10 / 0.22 / 0.45 / 0.60 GeV and one-year errors on the deformation coefficient of 1.6% / 5.7% / 104% / 540%. The aperture itself is measured, not assumed: an intact ^{6}Li through the ePIC far-forward geometry (`tools/fullsim/ion_gun_hepmc.py`, 84 points in p_T × azimuth) clears the pots at $|\theta_x| \gtrsim 2.0$ / 1.35 / 1.03 mrad in the 5×41 / 10×100 / 18×275 optics — p_T = 0.25 / 0.41 / 0.85 GeV — against $|\theta_y| \gtrsim 1.8$–3 mrad, an α at the same rigidity clearing at the same angle, so this is optics, not species: the beamline images an IP angle onto the pot plane with a horizontal lever of 30.6 m against a vertical few metres. The aperture is tighter than the 10σ envelope at every energy and inverts the assumed aspect, so the fake ⟨cos 2β⟩ changes sign; a track must clear both, and the tagged fraction at B = 50 GeV^{-2} falls to 9.8×10^{-7} / 7.7×10^{-16} / 1.9×10^{-17} at 40.8 / 99.5 / 137.5 GeV/u (`nearbeam_aperture_scan.py`, γ-matched energies), against 7.5×10^{-2} / 1.4×10^{-5} / 2.0×10^{-9} if the horizontal approach were the 0.727 mrad of the proton-derived 10σ envelope. The scan is one event per point in 30° azimuthal steps, no beam envelope, September-2024 `epic-main` (plans/04 #20). The alternative is the IR-8 secondary focus, which reaches $p_T \approx 0$ [10] and whose ^{6}Li efficiency is our interpolation of the published d / ^{3}He / ^{4}He / ^{7}Li values.

7. **Resolution and unfolding.** The divergence smearing moves the effective slope from B to $1/(1/B + 2\delta p_T^2)$ (47.7 here) and smears |t| by $2p_T\delta p_T \approx 0.01$ GeV², small against the 0.04–0.2 GeV² window (Figure 2d); the t-binning migration and the acceptance edge are unfolded with the same emulation, and t_{min} is a known bias of 3×10^{-3} GeV² at $x_P = 0.01$ once x_P is known from M_X.



Schematic 3 — the coherent channel. Top: the far-forward measurement of the intact recoil, visible only when its transverse momentum carries it outside the beam envelope at the Roman Pots, and the vetoes that define exclusivity. Bottom left: the two physical azimuths relative to the alignment axis and the harmonic each mechanism modulates. Bottom right: the rectangular cutout, whose symmetry axes coincide with the spin axis.

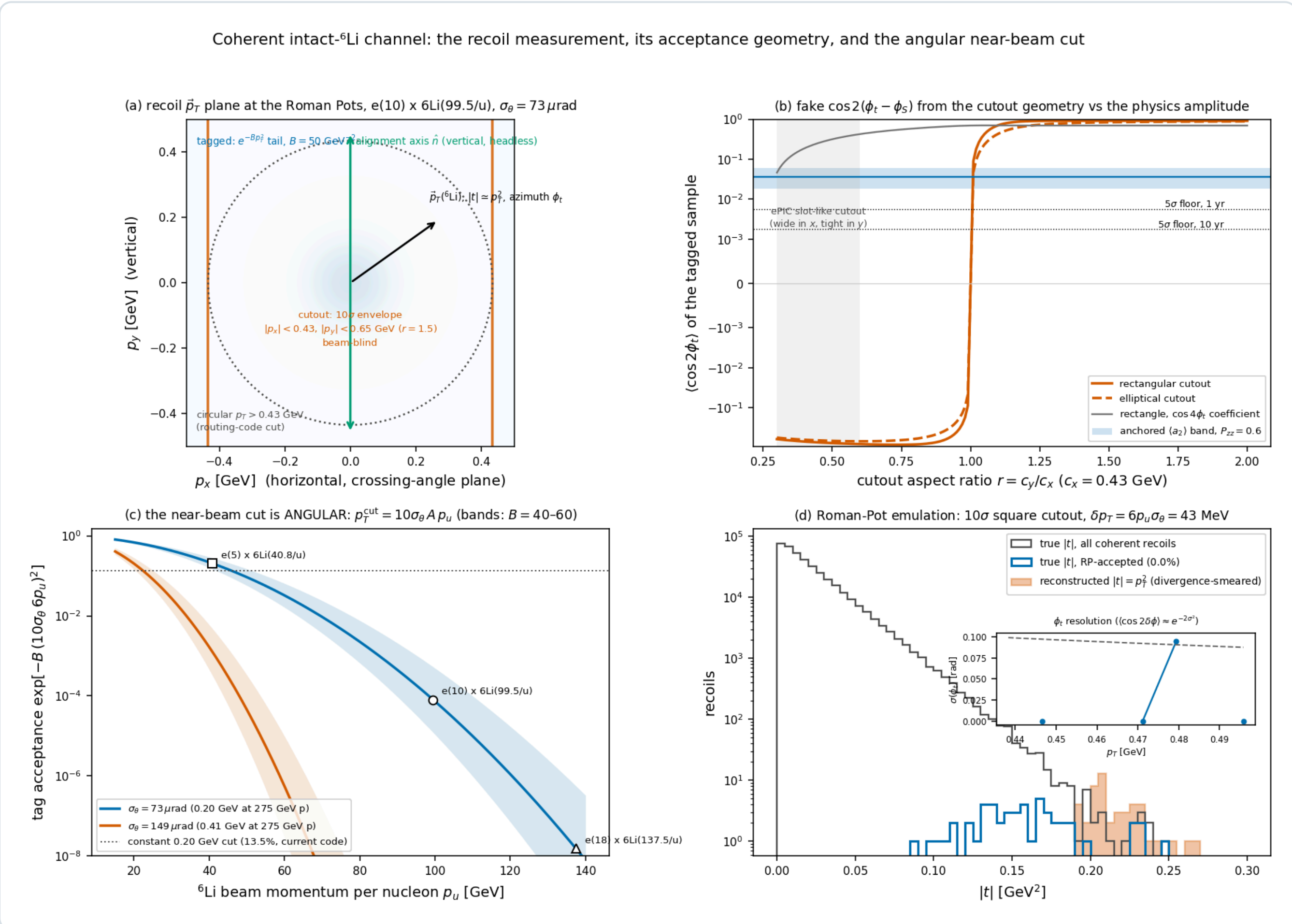


Figure 2 — the coherent measurement quantified. (a) The recoil transverse-momentum plane: the exponential coherent spectrum, the rectangular 10σ cutout (aspect ratio 1.5 drawn) against the circular cut of the routing code, and the vertical alignment axis. **(b)** ⟨cos 2φ_t⟩ of the tagged sample versus cutout aspect ratio for rectangular and elliptical cutouts, with the cos 4φ_t coefficient of the square, the anchored deformation band and the 5σ statistical floors of the 1- and 10-year programs. **(c)** Tag acceptance versus beam momentum per nucleon for the two proton-derived divergences, with the B = 40–60 GeV⁻² bands, the three beam configurations and the constant-cut value of the projection. **(d)** The Roman-Pot emulation (`reco.rp_measure`): true and accepted |t| spectra, the reconstructed |t| = p_T^2 after divergence smearing, and (inset) the φ_t resolution.

5 · What the simulation does

The measurement layer is `polligen/reco.py` and `polligen/hfs.py` [11], composed into the pseudo-experiments of §7 by `polligen/recopseudo.py` ; behind it sit the generator kernel and the spin bookkeeping (`bookkeeping.SpinCategory` , `RunPlan` , `tensor_flip_plan`). Table 4 walks the chain of §§3–4 step by step.

STEP	WHAT AN EXPERIMENT DOES	WHAT THE CHAIN DOES	STATUS
Spin bookkeeping, luminosity, polarimetry	Bunch pattern, L_f, P_{zz} per fill with a 3% scale	Spin categories with luminosity shares, relative-luminosity offsets, polarimetry smearing; the m = ±1-rich / m = 0-rich pattern of the estimator	**done**
e′ four-vector, crossing angle, head-on frame	Track + cluster → k′ in the lab; head-on transformation	`electron_fourvector` , `head_on_to_lab` , `lab_to_head_on` , applied per event	**done**
e′ resolution	Calorimeter energy, track angles	`emcal_resolution` (backward spec or the Yellow Report η table), `tracking_resolution` , angular resolution, energy-scale nuisance	**done, placeholder angles**
Electron ID and efficiency	E/p, shower shape, charge-symmetric subtraction	$\varepsilon_{eID}(\eta)$ on the ATHENA/ECCE anchors, with an η tilt as a nuisance	**no purity or background model**
Kinematic reconstruction	Mixed method at low y from the hadronic final state	Electron, JB, Σ, DA and mixed, on the generated final state or the parametrized stand-in	**done**
Hadronic final state and hadron-side detection	All tracks and clusters except e′, through the detector	`hfs.HadronResponse` (tracker coverage and thresholds, EMCal/HCal, calorimeter noise) on a PYTHIA 8 sample of 8 M events over the three beam configurations, with an energy-scale nuisance and a beam-energy guard	**done**
The physics azimuth	Covariant φ_S from k, k′, P, S	`azimuth_wrt_lepton_plane` , verified against an explicit collinear-frame construction to 10^{-13}	**done**
Reconstructed (x, Q^2) binning and migration	Reco bins, migration, unfolding	Purity, efficiency and reco-bin amplitude per bin; `fold` and `fold_shape_fit` per Q^2 slice (Table 5)	**done**
Spin-state-sorted ratio estimator	R(φ′) of m = ±1-rich against m = 0-rich fills, fitted with sin 2φ′	`spin_state_ratio` , `harmonic_ratio_fit` and its two-azimuth form, with the per-fill efficiency the ratio cannot cancel	**done**
Full-luminosity pseudo-data	—	Exact expected counts per φ′ bin, truth and reco level, Poisson-fluctuated	**done**
Amplitude → Δ	Kinematic conversion with measured F_2 and R; unfolding	The model bin-centering factor, or the folded shape fit with its shape, response-MC and prior-spread errors	**R and K from the model; F_2 cancels**
Radiative corrections	Unpolarized RC on the yields; tensor RC	—	**not modelled**
Coherent: recoil four-vector and t	t from the recoil, x_P from M_X	`recoil_fourvector` (exact light-cone), `t_from_fourvectors`	**x_P sampled, never reconstructed**
Coherent: Roman-Pot measurement	Transport, divergence smearing, cutout geometry, angular cut	`rp_measure` with the angular envelope *and* the aperture measured in the ePIC geometry, whichever binds; `rp_hole_acceptance` for any cutout; `coherent.py` keeps the constant p_T cut for the truth-level projections	**done**
Coherent: two-azimuth harmonic analysis	R(α, β) per (x_P, t, Q^2) bin	`harmonic_ratio_fit_2d` on the acceptance-weighted template basis, with the sin null columns and the assumed u_1, u_2 and luminosity as fit arguments — the per-fill cutout	**no e′ response; basis fails on the measured aperture**

		is a *generation* argument the fit never sees; α integrated analytically, one |t| bin at a time, no x_P or Q^2 binning	
Coherent: exclusivity and vetoes	OMD/ZDC/B0 vetoes, rapidity gap, Z-ID, |t| fit	`coherent.veto_table` bookkeeping	**no event-level breakup model**
Full simulation, far forward	Geant4 acceptance, EICrecon transport	Single-particle and intact-^{6}Li scans through the ePIC geometry, hit level (`tools/fullsim`)	**hit level; one event per point, no envelope, 2024 geometry**

Table 4 — the chain, step by step. "Done" means the step is modelled and tested, not that its inputs are final. Three inputs limit numbers in this note: the unsourced track angular table and the ePIC calorimeter noise floor (§3; plans/04 #21), and the Roman-Pot aperture, measured on a September-2024 `epic-main` whose pot stations have since moved once (plans/04 #20). Item-by-item status is plans/08; the external inputs are plans/04.

ITEM	WHAT WAS WRONG OR MISSING	WHAT CLOSED IT	WHAT IT PRODUCED
Hadronic final state	A 25% Gaussian stand-in for the hadronic y, then a toy fragmentation	A PYTHIA 8 production, 8 M events over the three beam configurations, through the same hadron-side response	$\delta y/y$ = 0.32 / 0.22 / 0.29 / 0.15 at the four mid-configuration sweet spots (0.38–0.11 at the low configuration's own spots, 0.23–0.18 at the top's; the $x \approx 0.14$ bins are best resolved at the low one), against a toy fragmentation that was optimistic everywhere; purity 43–69% against the Gaussian's 63–69%, which Table 2 and money plot 5R still run on — the PYTHIA y is the `--y-source hfs` cross-check
Roman-Pot cutout geometry	A slot orientation taken from a conference slide, with the acceptance an assumption	An intact ^{6}Li shot through the ePIC far-forward geometry in p_T and azimuth (84 points, one event each, September-2024 `epic-main`)	The aperture opens *horizontally*: $\|\theta_x\| \gtrsim 2.0$ / 1.35 / 1.03 mrad against $\|\theta_y\| \gtrsim 3.0$ / 3.0 / 2.3 in the 5×41 / 10×100 / 18×275 optics, the opposite aspect to the assumed slot, so the acceptance-induced $\langle \cos 2\beta \rangle$ about the vertical spin axis is large and negative (−0.71 at a 0.73 mrad horizontal approach), and at the γ-matched energies the measured aperture alone leaves no accepted recoil in the binned |t| window at any configuration (3×10^{-6} of the coherent yield at 5 × 41, all above |t| = 0.25 GeV²)
Bin-centering factor	K evaluated with the same Δ model that generated the pseudo-data	`fold` and `fold_shape_fit` : the amplitude is exactly linear in Δ, so the shape can be fitted *through* the response	Residual bias with a wrong prior up to 29% → at most 6% over the plotted bins of the three Q^2 slices and both wrong priors; sweet-spot model dependence (−4.1, +8.0, −5.5, +4.9)% → (−1.5, −1.8, −0.3, +0.5)% for the moment_B prior and (+8.6, +2.5, +7.3, +3.2)% → (+4.6, +3.0, +1.8, −0.9)% for the toy. `--unfold model` is still the default, so 7R still sits on the injected curve
The systematic the ratio cannot cancel	One $\varepsilon(\varphi')$ common to every fill, so only the case that cancels could be exercised	A per-fill acceptance, and the analytic bias $\Sigma_f\, l_f(P_f - \bar{P})e_f/\sigma_P^2$	$(\varepsilon_2^+ - \varepsilon_2^0)/(P_+ - P_0)$ at *any* luminosity split, so a lopsided run plan buys nothing; 10^{-3} fakes 5.6×10^{-4}, 1.9–6.0 one-year errors over the four spots
$R = \sigma_L/\sigma_T$	A placeholder of the right magnitude, and in the fast-simulation line a fit clipped to R = 1 over the region carrying the sensitivity	The published SLAC/E143 R1998 fit, all three functional forms, behind the `r_func` hook — transcribed and tested, but not the default	Of the two structure-function inputs only R moves the answer: F_2 cancels to 2×10^{-16}, and switching to R1998 would move Δ/F_1 by +17.8 / +18.5 / +7.6 / −4.4% at the four sweet spots; every number here is still the placeholder R
Coherent null test	The coherent fit was cos-only and had no null test at all	sin 2α, sin 2β and sin(α + β) columns, exactly forbidden by reflection symmetry	A spin-axis error shows in both azimuths, a pot roll in β alone; the roll is resolved to 5–8 mrad in year one in the three |t| bins that carry the statistics and 16 mrad in the highest, and Table 3 is unchanged with the columns on
Assumed u_1, u_2	The same unpolarized harmonics were	An independent assumed value on the fit side	δu_2 reaches a_e at *first* order through $a_t \delta u_2 \langle \cos 2\beta \rangle$: a ZEUS-1σ error moves a_e by 20% in the lowest |t| bin,

	handed to the generator and to the fit		rising to 35% in the highest as a_t grows
Detector nuisances	Argued rather than measured	A scan holding the response fixed and varying one thing, with common random numbers	Electron energy scale ±1% → 0.3–0.9% on $\hat{\Delta}$; hadronic-resolution mismatch 0.5–1.4%; a 1% residual hadronic-scale error on the calibrated PYTHIA y (`--hfs-scale`) 0.2–0.7%; ε_{eID} η tilt ≤ 0.03% — all below the 4–9% model dependence of an *assumed* K, though the first two exceed the 0.2–1.2% left by the folded bin-centering fit above
The near-beam cut versus the optics	Two points quoted for a curve the letter needs, and the cut applied as a momentum threshold to a nucleus	An analytic acceptance curve against the envelope, plus a full-response scan of acceptance and reach at six cuts, with the measured aperture marked on both	Through the full response, tagged fraction 32% / 3.0% / 4×10^{-5} / 2×10^{-7} and $\delta a_t/a_t$ = 1.6% / 5.7% / 104% / 540% at envelopes of 0.10 / 0.22 / 0.45 / 0.60 GeV — 6× and 113× the analytic curve at the two largest cuts, where the divergence smearing flattens the slope; the measured aperture is 1.4–2.8× the envelope, so the envelope never binds
η-dependent calorimetry	One backward specification everywhere	The Yellow Report table per η region, off by default	≤ 0.1% on Δ at all four sweet spots

Table 5 — the audit items, and what closed them. Every entry is a measurement made with the chain above. What remains open is in §6, and the item-by-item record with the numbers that did not fit here is plans/08.

6 · What is left to build

A harmonic basis that survives the strongly anisotropic measured aperture — fewer columns, wider bins, or |t| re-binned inside the window the cutout leaves — and money plot 6R re-derived on it; the event-level α + d breakup behind the two-component |t| fit, which needs BeAGLE for A = 6 (plans/06 carries the budget); and the qualified rows of Table 4. Item by item that is plans/08 §8.4, and the external inputs it waits on are plans/04.

7 · Reconstructed-level closure: money plots 5R, 7R and 6R

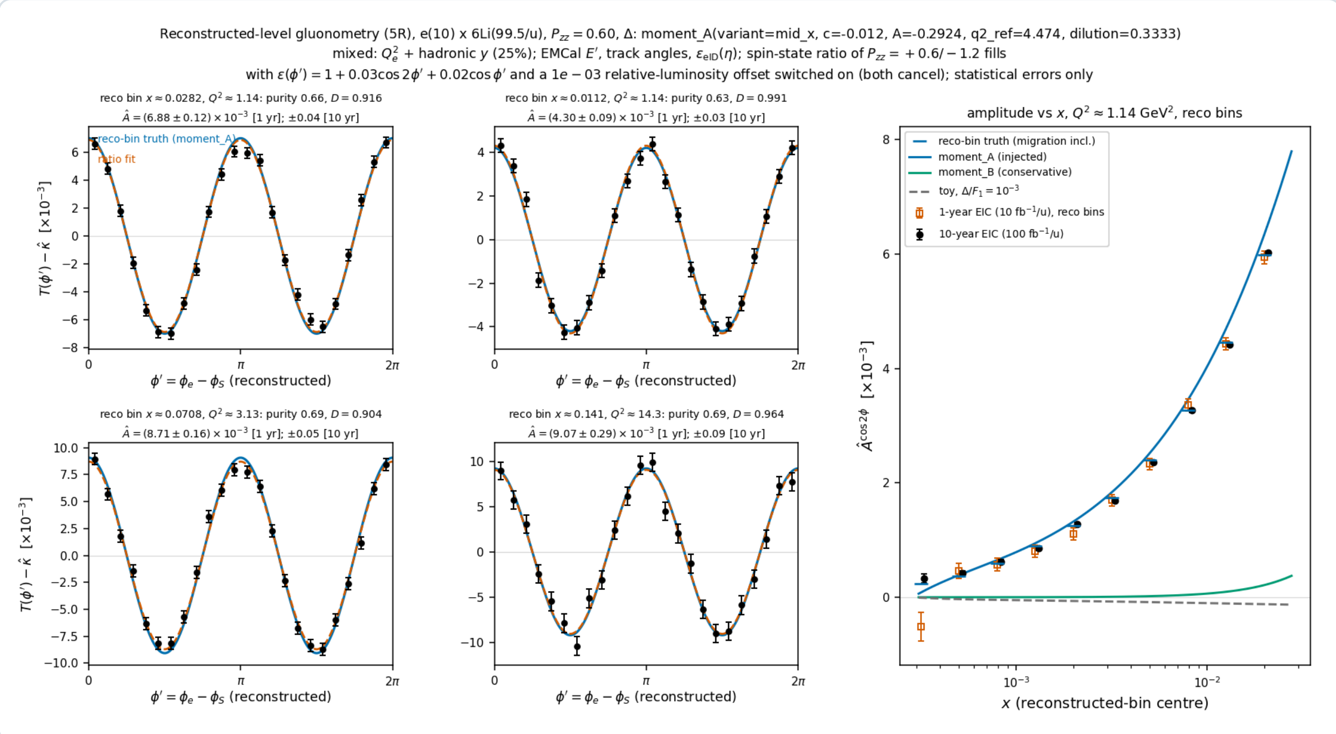
The chain of §§3–4 is composed in `polligen/recopseudo.py`. The inclusive response is importance-sampled: every accepted sampler cell (60×45 grid, generator cuts loosened to y ≥ 0.004, $Q^2 \geq 0.7$ GeV², |η| ≤ 3.8, E′ ≥ 0.3 GeV so that events can migrate *into* the analysis window) receives 400 pseudo-events of weight σ_c/n_{kept}, n_{kept} counting those inside the generator acceptance so that the cell rate is conserved. Each passes through the lab frame (25 mrad crossing), Gaussian smearing of E′ (2%/√E ⊕ 1%) and of the track direction (1–5 mrad by η), the head-on transformation, the electron-method Q^2, a hadronic y — the documented 25% stand-in [18,19] or, with `--y-source hfs`, the PYTHIA 8 final state through the hadron-side response — the mixed-method x, the analysis cuts on reconstructed quantities, $\varepsilon_{eID}(\eta)$, and the covariant φ′. The expected φ′ counts of a reconstructed bin are exact bin integrals of $den_f + P_f A \cos 2\varphi'$ over the events landing in it, Poisson-fluctuated at the 1- and 10-year luminosities for the two fills of the m = ±1-rich / m = 0-rich pattern (`bookkeeping.tensor_flip_plan`), with $\varepsilon(\varphi') = 1 + 0.03 \cos 2\varphi' + 0.02 \cos \varphi'$ and a 10^{-3} relative-luminosity offset switched on. The estimator is the spin-state ratio fit of §3 with the sin 2φ′ term; Δ follows with the MC bin-centering factor $K = \Delta_{model}(x_c, Q^2_c)/A_{model,\ reco\ bin}$, migration included.

SWEET SPOT (RECO BIN)	N, 1 YR (BOTH FILLS)	PURITY	EFFICIENCY	$D = A_{RECO\ BIN}/A_{TRUE\ BIN}$	$\hat{A} \pm \Delta\hat{A}$ (1 YR) [10^{-3}]	RECO-BIN TRUTH	$\Delta\hat{A}$ (10 YR)	ΔA OF MONEY PLOT 5
1: x = 0.028, Q^2 = 1.14	1.9×10^{8}	0.66	0.42	0.916	6.88 ± 0.12	7.00	0.037	0.17
2: x = 0.011, Q^2 = 1.14	2.9×10^{8}	0.63	0.59	0.991	4.30 ± 0.09	4.20	0.029	0.14
3: x = 0.071, Q^2 =	9.4×10^{7}	0.69	0.38	0.904	8.71 ± 0.16	9.08	0.052	0.28

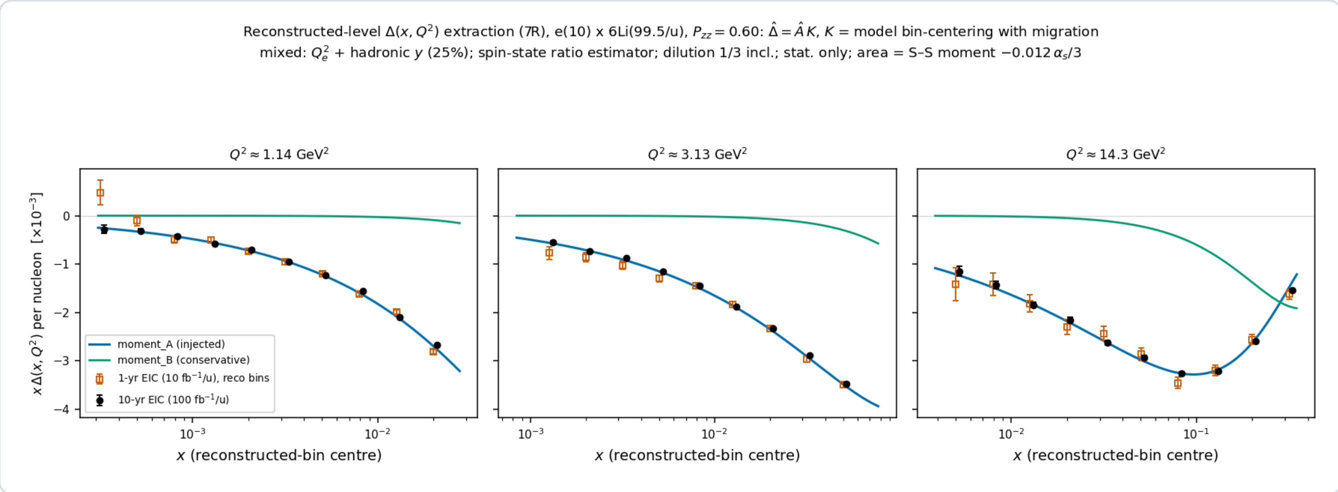
3.13								
4: x = 0.141, Q^2 = 14.3	2.9×10^7	0.69	0.66	0.964	9.07 ± 0.30	9.22	0.093	0.45

Table 2 — money plot 5R: the four sweet-spot super-bins in reconstructed (x, Q^2), mixed method with the documented 25% hadronic-y resolution [18,19]. Efficiency = fraction of the true-bin rate reconstructed in the bin after ε_{eID}; D = the rate- and ε_{eID}-weighted, φ′-diluted model amplitude of the events in the reconstructed bin over the rate-weighted amplitude of the true bin. The last column is the single-fill, truth-binned error of the projection report.

Three observations. (i) The fit is unbiased at the reconstructed level: every Â agrees with its reco-bin truth within about two standard errors while the acceptance harmonic and the luminosity offset are on, and the sin 2φ′ coefficients are consistent with zero. (ii) The two edge bins (spots 1 and 3 at y ≈ 0.01) lose about 60% of their events to the reconstructed y ≥ 0.01 cut and their amplitude reference shifts by 8–10%, the accepted half sitting at lower true x; K knows both, and the model's x-dependence is still resolved bin by bin (money plot 7R, best bins $\delta\Delta = 2.4\times10^{-3}$ at Q^2 = 1.14 and 1.2×10^{-3} at 3.13 GeV² in year 1, purities 0.54–0.57). (iii) The errors are *smaller* than the truth-binned single-fill projection, 0.59–0.69 of the money-plot-5 values, because the 1.5× gain of the m = 0-rich fill outweighs the reconstruction efficiency of 0.38–0.66 — the number to carry into plans/07 §7.1, and it presumes the source delivers m = 0-rich bunches at $|P_{zz}| = 1.2$, the same purity as the ±1 fill at 0.6.



Money plot 5R — inclusive gluonometry at the reconstructed level (e10 × ⁶Li99.5, P_{zz} = 0.6, moment_A, mixed method with 25% hadronic y, EMCal energy, track angles, $\varepsilon_{eID}(\eta)$, spin-state ratio of the +0.6/−1.2 fills with the acceptance harmonic and luminosity offset on). Left: the acceptance-free modulation $T(\varphi') - \hat{\kappa}$ per unit P_{zz} from the ratio, with the reco-bin truth (blue) and the fit (vermillion); annotations give purity, D and the 1-/10-year errors. Right: the amplitude in reconstructed x bins along the Q^2 = 1.14 GeV² slice against the reco-bin truth (blue ticks) and the model curves.



Money plot 7R — Δ(x,Q²) from reconstructed bins. xΔ points at the three sweet-spot Q² slices from the same pseudo-measurements, converted with the MC bin-centering factor K (model-derived, as in money plot 7, so the points sit on the injected curve by construction and carry the statistical errors of the reconstructed analysis); 1-year (open) and 10-year (filled) draws are independent.

The bin-centering factor, fitted rather than assumed. The cos 2φ′ amplitude is *exactly* linear in Δ, so `RecoResponse.fold` is the response as an exact linear operator and `fold_shape_fit` fits a Δ(x) shape through it per Q² slice on a bounded grid (`--unfold folded`; `--unfold model` stays the default and reproduces every published 7R number bit for bit). Correcting with either of two deliberately wrong priors, the residual bias over the plotted bins falls from 22% / 24% / 99% to at most 4.7% / 7.1% / 6.9% in the three slices, and the model dependence at the four sweet spots from (−5.0, +8.5, −9.3, +6.3)% to (−0.9, −1.2, −0.2, +0.3)% with the moment_B prior and from (+9.7, +2.5, +10.6, +2.8)% to (+4.7, +2.8, +2.2, −0.9)% with the toy. The 7R bar then carries four terms: the statistics, the shape fit, the response MC (0.10–1.00% per plotted bin, median 0.29%; 0.32 / 0.27 / 0.21 / 0.11% at the four spots) and — the dominant one — the spread over the priors the tilt family cannot absorb, measured to cover the residual bias at every plotted bin of every slice.

Cross-check with the hadronic final state. Rerunning 5R with the hadronic y from the PYTHIA 8 final state through the hadron-side response (Σ method, 50 MeV noise; `money_cos2phi_reco.py --y-source hfs --hfs-sample …`) gives purities 0.43 / 0.54 / 0.47 / 0.69, efficiencies 0.26 / 0.48 / 0.26 / 0.68, D = 0.79 / 0.85 / 0.82 / 0.95 and $\delta\hat{A}$ = 1.2 / 1.0 / 1.6 / 2.9 × 10^{-4}, the fit closing on the response-folded truth; the median captured Σ fraction is 0.92 (16–84%: 0.79–0.99) over the whole analysis acceptance, dominated by high y, and 0.72–0.87 at the four spots. The errors are the Table 2 errors; what moves is the purity and the dilution, because the real final state is only partly captured — 69–85% of Σ at the four spots through the response, the rest escaping forward beyond the calorimeters or lying below threshold (§3, Table 1b) — and these numbers carry that scale bias *uncalibrated*: the reconstructed-bin amplitude sits 5–21% below the true-bin value rather than 1–10%, so more of the answer passes through the bin-centering factor. With the analysis's own hadronic-scale calibration — the library's rate-weighted mean captured fraction per (x, Q²) cell, `--hfs-calibrate` — the purities are 0.52 / 0.59 / 0.59 / 0.76, the efficiencies 0.35 / 0.56 / 0.34 / 0.73 and D = 1.01 / 1.03 / 0.95 / 0.98 at unchanged errors (7R: δΔ = 2.2 / 1.1 / 0.3 × 10^{-3} at purities 0.44–0.57), so the uncalibrated case is what the scale bias costs on top of its event-by-event fluctuation and the noise; a residual scale error of ±2% then moves $\hat{A}$ by 0.3–1.5%, i.e. 0.2–0.7% per per cent. It stands as long as the calorimeter noise floor is ≈ 50 MeV, degrades in proportion to it, and the x ≈ 0.14 bins are better resolved at the low-energy configuration (§3).

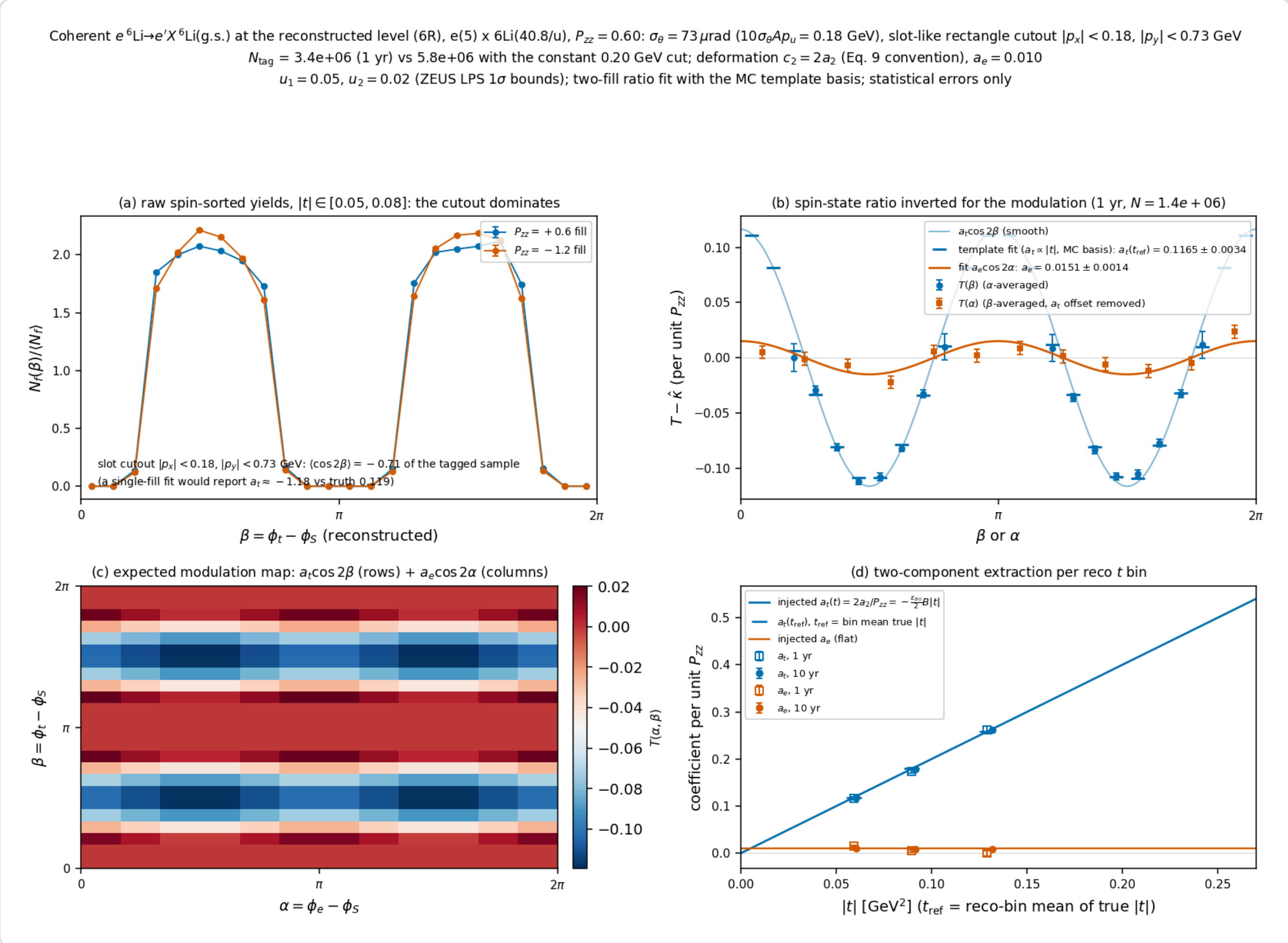
Coherent channel (money plot 6R). The coherent response samples recoils from the exponential spectrum with $x_P \in [10^{-3}, 10^{-2}]$, builds the exact four-vector and emulates the pots of §4 at the low-energy configuration, e(5) × ⁶Li(40.8 GeV/u): the measured aperture in the vertical (3.0 mrad) with a 0.73 mrad horizontal near-beam approach — the 10σ envelope of the proton-derived 73 μrad, standing in for the tagging optics of the projection report, since with the measured aperture and any published divergence no accepted recoil is left in the binned window at any configuration (below) — i.e. a cutout $|p_x| < 0.18$, $|p_y| < 0.73$ GeV, with divergence smearing. The expected counts per fill are exact α-integrals per event of $1 + u_1\cos(\alpha-\beta) + u_2\cos 2(\alpha-\beta) + P_f[a_e\cos 2\alpha + a_t(t)\cos 2\beta + a_m\cos(\alpha+\beta)]$, with the *true* β and t inside and the reconstructed β and $|t| = p_T^2$ for the binning. Three lessons. The anchor's convention: the deuteron paper's text fixes

$d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$ (its Eq. 9), so the cos 2β coefficient is $2a_2$, twice what money plot 6 injects ($c_2 = -(P_{zz}/2)\varepsilon_{B0}B|t|$ = 0.072 at ⟨|t|⟩ = 0.06 GeV², P_{zz} = 0.6; `coherent.cos2phi_coefficient_deformation`). The basis: the cutout leaves the in-bin β distribution far from uniform and correlates β with |t| within a reco t bin, the blind directions holding the larger-|t| recoils, so the fit uses the acceptance-weighted basis from the response — ⟨cos 2β⟩, ⟨cos β⟩, ⟨sin β⟩ per bin plus the template $\langle(t/t_{ref})\cos 2\beta\rangle$ of the assumed linear shape, plans/06's two-component fit $a_2(t) = c_{def}|t| + a_g$ in MC-response form; the analytic uniform-bin basis biases the deformation coefficient by 3–6% in the lowest t bin (1.25-aspect cutout), the template closes it. The geometry: the measured aperture is tight horizontally and wide vertically, so the accepted recoils cluster near β ≈ ±π/2 (⟨cos 2β⟩ = −0.71), where cos 2β hardly varies; the anisotropy costs the deformation harmonic statistics — a square cutout of similar acceptance would do about three times better — while a_e is untouched in principle, though at this aperture the template basis leaves it a residual bias of the order of the one-year error (Table 3). The pot geometry is a design lever for the deformation measurement and, to that order, irrelevant for the exotic-glue one.

The measured aperture alone. Without the near-beam approach the measured aperture (§4) leaves no accepted recoil in the binned |t| window at any of the three configurations (`money_cos2phi_coherent_reco.py --rp-aperture measured --cut-scale-x 1.0`): at 5 × 41 the accepted fraction is 3×10^{-6} and lies entirely above |t| = 0.25 GeV², at 10 × 100 and 18 × 275 it is zero. Before the beam energies were corrected (plans/10) the same chain at the then-assumed 20.5 GeV/u had kept two |t| bins with 1.4% acceptance; that result is superseded. The 0.73 mrad approach of Table 3 recovers 7.8% at 5 × 41 with three of the four |t| bins (the highest is rank-deficient) and ⟨cos 2β⟩ = −0.71; the same approach at 10 × 100 keeps 9×10^{-5} and one bin. The deformation measurement is therefore a low-energy one and exists only with a near-beam tagging capability — a tagging optics with pots that follow, or the IR-8 secondary focus — which is the requirement the projection report states.

RECO |T| BIN [GEV²]	N_{TAG}, 1 YR	$A_T(T_{REF})$ INJECTED	FIT, 1 YR	FIT, 10 YR	A_E FIT, 1 YR (INJ. 0.010)	A_E FIT, 10 YR
0.05–0.08	1.4×10^6	0.119	0.1165 ± 0.0034	0.1176 ± 0.0011	0.0151 ± 0.0014	0.0111 ± 0.0004
0.08–0.12	4.6×10^5	0.180	0.174 ± 0.004	0.1785 ± 0.0013	0.0051 ± 0.0025	0.0095 ± 0.0008
0.12–0.17	8.3×10^4	0.258	0.262 ± 0.008	0.2612 ± 0.0026	0.0006 ± 0.0058	0.0089 ± 0.0018
0.17–0.25	dropped: 111 accepted response recoils, the seven-column harmonic design is rank-deficient at this aperture					

Table 3 — money plot 6R: template fit of the two-azimuth spin-state ratio per reconstructed |t| bin (coefficients per unit P_{zz}; ε_{B0} = −0.08, B = 50 GeV⁻²; u_1 = 0.05, u_2 = 0.02 at the ZEUS 1σ bounds [13]; $N_{tag} = 3.4\times10^6$ in year 1 at e(5) × ⁶Li(40.8 GeV/u) for the measured vertical aperture and a 0.73 mrad horizontal approach, against 5.8×10^6 for the constant 0.20 GeV cut at this configuration). The cutout's fake ⟨cos 2β⟩ is −0.71 — a single-fill fit would report $a_t \approx -1.2$ for a true 0.12 — and the ratio removes it entirely; the a_e of the lowest bin sits 3.6σ (one year) and 2.8σ (ten years) above the injected value, and a second pseudo-experiment (`nearbeam_reach_gain.py`) returns 0.0102 ± 0.0014 there, so the template basis carries a residual bias of the order of the one-year error at this aperture. The fit carries the three parity-forbidden sin harmonics as a null test (§4): all twelve coefficients are consistent with zero, and their errors resolve a Roman-Pot azimuthal roll in year one to 5–8 mrad in the three |t| bins that carry the statistics and 16 mrad in the highest.



Money plot 6R — the coherent channel at the reconstructed level. (a) Raw spin-sorted yields versus the reconstructed recoil azimuth β in the lowest |t| bin: the cutout shape dominates both fills identically (the recoils near β ≈ ±π/2 clear the horizontal aperture). **(b)** The modulations T(β) and T(α) from the spin-state ratio (T(α) with the a_t offset removed), the template-fit prediction per β bin (ticks) and the smooth $a_t\cos 2\beta$ and $a_e\cos 2\alpha$ curves. **(c)** The expected (noise-free) modulation map in (α, β): rows from the deformation term, columns from the gluon-transversity term. **(d)** $a_t(t_{\rm ref})$ and a_e per reco |t| bin (1 yr open, 10 yr filled) against the injected curves.

7.1 · Known limits and systematics

Known limits of these numbers. The hadronic y of Table 2 is a parametrized 25% Gaussian — an absolute 50–125 MeV on Σ_h = 0.2–0.5 GeV at the sweet spots — so it rests on the real detector's floor at that Σ_h; u_1 and u_2 are generated at the ZEUS 1σ bounds [13], and the slot dimensions and the φ′-efficiency harmonics are illustrative magnitudes; radiative corrections and backgrounds are absent, and the 6R sample is background-free and Z-identified by assumption; the 7R points are centered with the injected model unless the folded fit is used; and the template assumes the linear t-shape it fits, so a different shape is a model systematic to be varied. The fitted coefficients refer to the unsmeared truth: the α-integrals are analytic per event and the β smearing is absorbed in the MC basis. The sign of the deformation term is positive for P_{zz} > 0 and ε_{B0} < 0, the predicted flip relative to the deuteron, with the Eq. (9) convention read from the paper's own text.

Detector nuisances, measured. Each is switched on in the response and the amplitude converted with the nominal K, so the quoted number is the shift of $\hat{\Delta}$ (`money_cos2phi_reco.py --syst-scan`, common random numbers; the response's own MC noise floor is 0.13–0.21% at the four spots). A 1% electron energy-scale error moves $\hat{\Delta}$ by 0.2–1.4%, the larger of the two energy levers since d ln x/d ln E′ = 2 − y for the Σ method against −(1 − y) for the hadronic scale — the Gaussian y stand-in never sees E′ and gives exactly 1, understating this systematic by about two. Generating at a 30% hadronic resolution and correcting with the 25% response shifts $\hat{\Delta}$ by 0.5–1.3%. An η slope of 0.05 per unit on $\varepsilon_{\rm eID}$ gives 0.02%: the estimator is a weighted mean of per-event amplitudes, so a smooth efficiency shape is almost exactly null and a flat one exactly null. Replacing the backward-endcap calorimeter specification by the Yellow Report η table changes nothing at spots 1–3, whose electrons are backward, and 0.5% at spot 4, whose η = −1.64 reaches the transition region. All four sit below the 2.5–11% model dependence of an assumed K (moment_A/moment_B/toy at the four spots), itself larger than the 10-year statistical errors unless K is fitted through the response (above).

The systematic the ratio cannot cancel (docs/code_review_2026-08-25.md §5.1). The ratio cancels only a *common* acceptance, and the φ′ efficiency of Table 2 is common to both fills. A difference $\delta\varepsilon_2$ of the cos 2φ′ harmonic between the m = ±1-rich and m = 0-rich samples enters $\hat{A}$ as $\delta\varepsilon_2/(P_+ - P_0)$ (`reco.fill_acceptance_bias` , verified against the pseudo-experiments): $\delta\varepsilon_2 = 10^{-3}$ fakes 5.6×10^{-4}, 5% of the sweet-spot amplitudes but 1.9–6.0 one-year statistical errors and 6–19 ten-year ones over the four spots. The coherent counterpart is *tighter*: a naive $\delta\langle\cos 2\beta\rangle/(P_+ - P_0)$ gives 1.3% of a_t for a 1% fill-to-fill change of the vertical envelope, but the fit gives +19% for a $\mathbf{10^{-3}}$ change and +169% for 1% — under the slot only half the β bins are live and the t-shape template is 99% anti-correlated with the constant there, so a shape perturbation is amplified about a hundredfold. Bunch-by-bunch alternation, or ≈ 10^{-4} envelope stability, is therefore a requirement of the measurement; failing that, the higher-|t| bins are far less exposed (+3.9% and +0.04% at the same 10^{-3} for |t| = 0.08–0.12 and 0.12–0.17 GeV²). The EMCal term 2%/√E ⊕ 1% is the backward specification applied at all η (the sweet-spot electrons are backward, the high-x bins of the amplitude panels are not). u_1, u_2 and the luminosity shares are exactly known in Table 3, but as fit arguments (`measure_coherent(u_coeffs_assumed=, lumi_assumed=, responses=)` , `money_cos2phi_coherent_reco.py --u2-assumed --rel-lumi-offset --envelope-split`), which is where the sensitivities above come from. The relative luminosity is second order in both channels: inclusively a 2% offset moves the amplitude by 0.7% and a 10^{-4} offset by 3×10^{-5}, and 10^{-3} moves a_t by 0.08%.

8 · What the chain assumes, and what it needs

Everything above rests on inputs that belong to the accelerator, to the detector, or to a measurement nobody has designed yet. This section is the consolidated list, split by **who owns each**, because the response differs: an accelerator input is something to ask for precisely; a detector requirement is something to hold ePIC to; and a property of *this* measurement is something we have to specify ourselves and then meet. Report 3 carries the machine and detector specification in full; this is what the chain actually consumes.

8.1 · What the chain assumes

INPUT	VALUE USED	STATUS	IF IT IS WRONG
Beam			
⁶Li energies	`40.8 / 99.5 / 137.5 GeV/u`	derived — γ-matched, then rigidity-capped (plans/10)	everything: p_ion sets every far-forward cut
σ_θ h/v, high acceptance	`220/380, 180/180, 92/92 µrad`	YR Table 10.1 + the species step	far-forward acceptance, exponentially
Δp/p	`7–10 ×10⁻⁴`	YR Tables 10.1/10.2; species-insensitive	little — no observable here uses it
P_zz (projections)	`0.60`	**conservative** — the ANL source target is ≥ 0.80	statistical error only; 0.80 would improve it 33%
tensor flip states	`+0.6 / −1.2, equal luminosity`	derived; the equal split is exactly optimal	statistical error
electron / ion vector polarization	`0.70 / 0.70`	YR assumption for physics studies	not used by the tensor observable
integrated luminosity	`10 / 100 fb⁻¹ per nucleon`	YR convention (1 yr / 10 yr)	statistical error
Detector			

EMCal σ_E/E	`2%/√E ⊕ 1%` `(backward)`	YR **requirement**, correctly sourced backward	$\delta Q^2/Q^2$ at the sweet spots
tracking σ_p/p	`√((a p)² + b²), η-piecewise`	YR **requirement** (Fig. 8.3 / Table 11.2) — *not* a placeholder	≈±20% against ePIC full simulation
tracking σ_θ	`3 mrad at the sweet spots`	**PLACEHOLDER** — no YR angular requirement exists	$\delta Q^2/Q^2$; but it is also the model's only stand-in for the 81–211 µrad beam-divergence floor, so the honest bracket is a factor ≲ 3
electron-ID efficiency	`0.70–0.95 by η`	constructed; sources disagree by ≈25 points in the barrel	**nothing** — a fill-independent efficiency cancels in the ratio
hadronic y resolution	`Σ method on the PYTHIA 8 HFS`	measured on 8 M events at the corrected energies	bin purity and the amplitude dilution D
hadron-side calorimetric reach and capture	`|η| ≤ 3.7 (tracker 3.5)`	response default; the ePIC nominal is 4.0	69–85% of Σ_h captured at the sweet spots through the response (§3, Table 1b): a scale bias on y_Σ that the pseudo-experiments carry uncalibrated (Table 2) or calibrate per cell (§7, `--hfs-calibrate`); 4.0 recovers a quarter of the forward loss with δy/y unchanged
Roman-Pot aperture	`2.00 / 1.35 / 1.03 mrad`	**measured in a superseded (Sept-2024) geometry**	coherent acceptance; re-measurement is plans/09 B1
Physics			
coherent t-slope B	`50 GeV⁻²`	model, ≈ $R^2/4$ for ^{6}Li	coherent acceptance, exponentially
nuisance harmonics u_1, u_2	`0.05, 0.02`	ZEUS LPS 1σ edge (NPB 816:1)	u_2 leaks into a_e at first order through ⟨cos 2β⟩
diffractive fraction f_0	`0.04`	model bracket (plans/06)	coherent rate, linearly
R = σ_L/σ_T	`toy 0.157–0.171`	the published R1998 fit is transcribed but not the default	Δ/F_1 by +17.8 / +18.5 / +7.6 / −4.4%

8.2 · The target specifications this measurement needs

Two of these are **ours to specify**, because they are properties of a measurement that does not exist yet and no EIC document addresses them: the tensor-polarization scale and the relative luminosity between tensor states. The rest are asks, with owners.

#	WHAT MUST BE ACHIEVED	TARGET	WHY THAT NUMBER	OWNER
T1	**Tensor-polarization scale, δP_zz/P_zz**	`≤ 5%`	a pure normalisation on Δ, propagating **1:1** (§8.3–8.4). 3% is reached on a stored beam (Nuclotron 2.1%, COSY/ANKE ≈4%) and on the HERMES storage cell; 8–10% is where solid-target work sits. The **dominant** systematic of the two we specify	**ours** + ANL source / BNL polarimetry
T2	**Relative luminosity between P_zz states**	`≤ 10⁻³ on L₁/L₂`	the bias is 1/3 per unit ratio error, so 10^{-3} costs 0.03% — two orders below T1. **Bunch-by-bunch alternation, if the source allows it, inherits the 10^{-4} RHIC achieves and makes this negligible.**	**ours** + C-AD / source
T3	Source tensor polarization	`P_zz ≥ 0.80`	the ANL ECRP target; the projections use 0.60, so meeting it improves the statistical error by 33%	ANL source
T4	Polarization survived to store	–	^{6}Li cannot use Siberian snakes (G = −0.178); 81 resonances to cross	C-AD / EPIOS

T5	Far-forward aperture, for the coherent tag	`10σ envelope, and a pot that reaches it`	the tag is $\exp(-B(10\sigma\_\theta \cdot A \cdot p\_u)^2)$; at published optics it does not survive (report 4 §7)	C-AD (β*) + ePIC FF (granularity)
T6	Charge identification in the pots	`α↔⁶Li fake ≲ 10⁻³`	^{6}Li → α + d puts both fragments at beam rigidity; four AC-LGAD planes reach it *if* EICROC's charge range does not clip at 9 MIP	ePIC FF / OMEGA-IJCLab
T7	Hadronic-method y resolution at y ≈ 0.01	`δy/y ≲ 0.3`	three of four sweet spots sit there and the electron method gives 0.9–2.3; the Σ method on PYTHIA gives 0.28–0.55	achieved in simulation; needs ePIC full sim
T8	A far-forward reconstruction that knows what lithium is	–	EICrecon selects its transfer matrix by matching the beam *proton* PDG code. **This is what the development is for, not a gap in the specification.**	**this programme**

8.3 · The two we specify, and what they cost

Corrected 2026-08-27. An earlier version of this section said a P_zz scale error propagates as $(1+d)^2 - 1 \approx 2d$. **It does not — it propagates 1:1.** The estimator's weights $w\_f = P\_f - \bar{P}$ are built from the *assumed* polarizations, so the ratio R carries one power of the assumed scale while σ_P^2 carries two, and one power cancels: **$\hat{A}/A$ = P_zz(true)/P_zz(assumed), exactly.** Verified to machine precision at every sweet spot and |t| bin. The quadratic is real but belongs to *reach*, not bias: $\delta A \propto 1/\sigma\_P$, so the luminosity needed at fixed δA goes as $1/P\_{zz}^2$. The published cost was 2× too pessimistic.

ΔP_ZZ/P_ZZ	AMPLITUDE SCALE ERROR	ERROR ON L₁/L₂	BIAS ON THE AMPLITUDE
2%	2.0%	10^{-4}	0.003%
5%	5.0%	10^{-3}	0.03%
10%	10.0%	10^{-2}	0.3%

Relative luminosity: to first order the bias is $-[(P_1+P_2)/(P_1-P_2)] \times \delta$, which for the flip plan's (+0.6, −1.2) is exactly **1/3 per unit error on the ratio** — measured 0.333 and stable to a few per cent out to $\delta = 10^{-2}$. **Mind the convention**: the scripts' `--rel-lumi-offset d` sets the assumed shares to [0.5(1+d), 0.5(1−d)] against equal truth, which is a *ratio* error of ≈2d, so the apparent coefficient there is ≈2/3. Both propagations are pinned as tests in `evgen/tests/test_recopseudo.py`.

8.4 · What δP_zz/P_zz should we target?

No EIC document states one, and **nothing exists for ^{6}Li at all** — EPIOS's proposed Li–Li CNI polarimeter is explicitly vector, and it says tensor polarimetry "requires specialized polarimeter development such as Lamb-shift-based systems or BRP-type polarimeters", both source-level devices. So the target has to be set from precedent on other species, and the distinction that matters is **target** versus **stored beam**.

PRECEDENT	ΔP_ZZ/P_ZZ	WHAT IT WAS	STATUS
Stored beams — where this programme lives			
JINR Nuclotron internal-target polarimeter	**2.1%**	P_zz = −1.469 ± 0.031, absolutely calibrated on ^{12}C(d,α)^{10}B* at 0° where the analyzing power is fixed by selection rules	measured (conference proceedings, table marked preliminary)
COSY/ANKE stored deuterons, 1170 MeV	**≈4%**	on both P_z and P_zz, from the ANKE analyzing-power transfer; *excludes* the EDDA absolute calibration, which adds 6–7%	measured
Targets — where most tensor precedent lives			
HERMES ABS deuterium storage cell	**3.0%**	the best tensor number in the literature; systematic-dominated by the *storage cell* (recombination, wall depolarization) — a mechanism a stored beam does not have	measured
… the same source's injected atomic beam	0.4%	before the cell	measured

NIKHEF/AmPS	4.9%	on the state *difference* ΔP_{zz}	measured
UVa solid polarized target	4.7–9.7%	RF-enhanced d-butanol	measured
VEPP-3	8.0%	total, stat ⊕ syst	measured
JLab E12-13-011 (b_1) budget	8.0%	"polarimetry"; 9.2% total scale on A_{zz}	requirement

> **The programme's stated assumption.** Take **$\delta P_{zz}/P_{zz}$ = 5%** as the working value and ≤ **5%** as the target specification (T1), with **3%** as the optimistic case that two independent experiments have already reached on a stored beam and on a storage cell, and **8–10%** as the conservative case that solid-target and lower-energy work sits at. On the corrected 1:1 propagation that is a **5% normalisation uncertainty on Δ**, with 3% and 10% as the bracket.
>
> **Relative luminosity: ≤ 10^{-3} on the ratio**, costing 0.03%. It is not a leading systematic at any plausible value, and if the source can alternate the tensor state bunch-by-bunch — the way RHIC alternates proton helicity, which is exactly what makes its 10^{-4} achievable — it stops mattering at all. **That is a design choice to make early and state, not a number to measure later.**
>
> **One caveat on the run plan.** The estimator sees the *lever arm*, not the scale: with per-state errors the bias is $(\delta_+ - \delta_0)/(P_+ - P_0)$, and the fractional weights are 1/3 and 2/3 for a $P_0 = -2P_+$ pair, independent of the luminosity split. That −2 ratio is a hardcoded idealisation in `bookkeeping.tensor_flip_plan` ("the same source purity") and is itself an assumption about the source, not a property of the measurement.

> **What this means for what we can claim.** A P_{zz} scale error multiplies every bin identically, so it is a pure **normalisation** on $\Delta(x,Q^2)$. *Every shape claim in this programme is immune to it* — the x and Q^2 dependence, the sign, the ratio between sweet spots, and the discrimination between the two Δ ansätze of §7 all survive a wrong P_{zz} scale untouched. What it moves is the **absolute magnitude** of Δ. Design the claims accordingly: lead with shape, quote magnitude with the P_{zz} scale attached.

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Appendix A · Revision history

DATE	REVISION
2026-08-27	Numbers restated at the γ-matched beam energies (^{6}Li at 40.8 / 99.5 / 137.5 GeV/u, plans/10) from the current scripts: the sweet spots (x = 0.011–0.14) and Table 2, the hadronic-y resolutions at the corrected spots (`hfs_resolution.py` now selects them per configuration instead of a constant), the bin-centering and nuisance scans, the R-model shifts. Money plot 6R and Table 3 re-derived: with the measured aperture and any published divergence no recoil survives at any configuration, so 6R is shown at the low-energy configuration with a 0.73 mrad near-beam approach; the pre-2026-08-27 measured-aperture result at the superseded 20.5 GeV/u is withdrawn. Companion papers: Reports 0 and 1 rewritten the same day. Later: §3 gains Figure 4 and Table 1b (`hfs_acceptance.py`) — where the hadronic $E - p_z$ sum goes at the four sweet spots: 78–91% within acceptance and above threshold, 69–85% captured through the response, 7–19% escaping forward beyond |η| = 3.7 — a scale bias on y_Σ that the library reproduces and the bin-centering factor absorbs, not the resolution driver; `HFSResponse(calibrate=True)` added as the analysis's own per-cell hadronic-scale calibration, with the calibrated 5R numbers in §7; the pre-correction "0.90 captured, 0.11 neutral hadrons" statement replaced.
2026-08-27 (later still)	**Corrected the δP_zz propagation, and set T1 from precedent.** The row above published $(1+d)^2 - 1$ = 4.0 / 10.3 / 21.0% at δP_zz/P_zz = 2 / 5 / 10%. **That is 2× too pessimistic: the propagation is 1:1**, so the costs are 2.0 / 5.0 / 10.0%. The estimator's weights are built from the *assumed* polarizations, so the ratio carries one power of the assumed scale against σ_P²'s two and one cancels — Â/A = P_zz(true)/P_zz(assumed) exactly, verified to machine precision. The quadratic belongs to *reach* (δA ∝ 1/σ_P) not to bias. The relative-luminosity coefficient was likewise wrong: it is **1/3 per unit error on the ratio**, not ≈1.4 × δ — the earlier number conflated the scripts' `--rel-lumi-offset` convention (assumed shares [0.5(1±d)], a ratio error of ≈2d) with the ratio error itself. The test that was supposed to pin the first checked only that σ_P² ∝ $(1+d)^2$ and never ran the estimator; both tests are rewritten to run it. New **§8.4** sets T1 = **≤ 5%** from precedent, since no EIC document states one and nothing exists for ^{6}Li at all: 2.1% (JINR Nuclotron) and ≈4% (COSY/ANKE) on stored deuteron beams, 3.0% on the HERMES storage cell, 8–10% for solid targets and the JLab b_1 budget.
2026-08-27 (later)	**New §8: what the chain assumes, and what it needs.** The consolidated list of every input the chain consumes, split by who owns each — accelerator, detector, physics model — with its status and what moves if it is wrong; then the target specifications the measurement needs, with owners. Two of those are **ours to specify**, because they are properties of a measurement that does not exist yet and no EIC document addresses them: **δP_zz/P_zz**, which is a pure normalisation, and the **relative luminosity between tensor states**, which is two orders less important. *(The propagation quoted in this revision was wrong by 2× and was corrected the same day — see the row below.)* Both propagations are pinned as tests. The consequence for what we claim: **every shape claim is immune to a P_zz scale error** — it moves only the absolute magnitude of Δ — so the claims should lead with shape and quote magnitude with the scale attached.

2026-08-27	**The beam energies were wrong at two of the three configurations, and everything conditioned on them is re-run.** The HSR and ESR must share a revolution period and the electrons are ultrarelativistic, so the hadron γ is fixed by the ring circumference: EIC ions are **γ-matched**, not rigidity-scaled. ^{6}Li therefore sits at **40.8 / 99.5 / 137.5 GeV/u**, not 20.5 / 50 / 137.5 (plans/10; the decisive check is that YR Table 10.2 lists gold at 41 GeV/u, not the 16.4 rigidity-scaling gives). The PYTHIA 8 hadronic final state was regenerated at the corrected energies for the two lower configurations — 8 M events, 80.4 M particles — and every figure and number derived from them re-run. What moved: Table 1's y values halve and δy/y at the mid configuration doubles (1.2 / 0.46 / 1.1 / 0.23 → 2.3 / 0.93 / 2.1 / 0.46), Σ-method δy/y goes 0.32 / 0.22 / 0.28 / 0.11 → 0.55 / 0.28 / 0.50 / 0.15 at mid and 0.21 / 0.12 / 0.17 / 0.10 → 0.28 / 0.21 / 0.24 / 0.11 at low, purity 0.40 / 0.53 / 0.44 / 0.73 → 0.43 / 0.54 / 0.47 / 0.69 and D 0.79 / 0.84 / 0.86 / 0.96 → 0.79 / 0.85 / 0.82 / 0.95. **The top configuration is rigidity-capped and did not move**, and neither did any number derived from it — which is the check that nothing else drifted. Finding (1) is strengthened, not weakened: the electron method is worse at the corrected energies, so x must come from the hadronic final state by a wider margin. Separately, the far-forward near-beam envelope is now per-configuration (`reco.sigma_theta_for`) rather than a single proton-derived 72.7 μrad, which is what plans/10 is about.
2026-08-24	First version: what is measured, the covariant azimuth and the head-on frame, the inclusive and coherent chains, the audit of the simulation and the recommendations (Figures 1–2, Schematics 1–3, `polligen.reco`). Same day: §7, the reconstructed-level pseudo-experiments of WP3 (money plots 5R, 7R, 6R from `polligen.recopseudo` , the acceptance-weighted template basis, the anchor's $2a_2$ convention).
2026-08-25	Revised against the reference copies in refs/: the ePIC Roman-Pot slot geometry and beam-effect resolution [17], the ZEUS bounds on the diffractive harmonics [13,14], the exact spin-1 harmonics at finite γ [12], the anchor's frame and convention [9]; the hadronic-y resolution set to the documented 25% [18–20] and 5R/7R rerun; the limits of §7.1 from the code review. Then the hadronic final state itself (WP3-HFS): `polligen.hfs` , the hadron-side response and the (x, Q^2) event library; §3, Figure 3 and the §7 cross-check.
2026-08-26	Three open items become measurements and one input moves a result. **The hadronic final state is PYTHIA 8**, 8 M events over the three beam configurations: §3, Figure 3 and finding (1) re-derived on it — δy/y = 0.55 / 0.28 / 0.50 / 0.15 at the sweet spots against the toy stand-in's 0.28 / 0.17 / 0.24 / 0.07, purity 40–73% against the 25% Gaussian's 64–68%. **The Roman-Pot aperture is measured** with an intact ^{6}Li through the ePIC geometry — it opens horizontally, the opposite aspect to the slot §4 assumed — and carried through the coherent chain (§4, §7). **The bin-centering factor has a fitted alternative**, `RecoResponse.fold` and `fold_shape_fit` (§7). **The published SLAC/E143 R1998 fit is transcribed for R** [21], behind `r_func` and not yet the default: F_2 cancels exactly and switching would move Δ/F_1 by +17.8 / +18.5 / +7.6 / −4.4% (§3). The $O(\gamma^2)$ b_1 leakage bound of §2 is corrected upward by ≈ 6.9× to the full Cosyn Eq. (17e) combination (plans/08 D2). Editorial: §5 becomes two tables — the chain step by step, and the audit items with what closed each — and §6 the work that is left; the item-by-item status is plans/08, and each validity limit stands beside the number it qualifies.

Conventions: head-on frame with the ion along +z and the electron along −z; θ′ is the scattered-electron angle from the electron beam; per-nucleon (x, y, s) throughout; whole-nucleus four-momenta where a mass matters. Beam divergences are the proton values implied by the documented Roman-Pot cuts (73 and 149 μrad); the lithium optics are undocumented (plans/04 #11). Built by `reports/build_report.py`; figures from the `evgen/scripts` named in the text; display mathematics typeset at build time.